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Supplemental Materials

for

Pseudomonas Isolation and Identification: An Introduction to the Challenges of Polyphasic Taxonomy

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Appendix 1: Materials and equipment list.

Student Supplies (provided at the bench)

- Loop
- Needle
- Bunsen Burner
- Microscope Slides and Cover Slips
- Clothes Pin
- Forceps
- Pipet Aids
- Tips for micropipettors, sterile
- Microcentrifuge tubes, sterile
- PCR tubes to fit the thermal cycler, sterile
- Racks to hold microcentrifuge and PCR tubes
- Labeling tape

Equipment

- Microscope (one per pair of students)
- Vortex mixer (one per four students)
- Microcentrifuge (one per 12 students)
- Micropipettors to cover volumes ranging from 2 μl to 1000 μl (one set per 2 students)
- Thermal cycler (capacity of one tube per student)
- Incubators at 4°C, 30°C, 42°C
- Shaking incubator or water bath set to 30°C and clamps to hold 250 ml flasks
- Anaerobic Gas-Pak jars with gas-generating sachet (optional, capacity of 12 plates)

Bacterial Media

All solid media are stable for months when stored at 4°C in sealed plastic bags.

Pseudomonas Minimal Medium (PMM)

This medium is prepared as a broth for the enrichment and as solid media for streak plates.

- Phosphate buffered basal salts solution, pH 6.8
  - 15 ml of 1 M NaH_2PO_4
  - 15 ml of 1 M K_2HPO_4
  - 970 mL deionized H_2O
  - Confirm that the pH of the mixed phosphate buffers is 6.8; adjust if needed.
  - Dissolve the following three components in the liter of phosphate buffer just prepared.
  - 0.5 g $MgSO_4$
  - 0.05 g $FeCl_3$
  - 0.005 g $CaCl_2$
  - Sterilize by autoclave.
- Carbon Sources: Will vary depending on experiment design. We use sodium benzoate, p-hydroxybenzoate, sodium succinate, histidine, arginine, trehalose, and penicillin G.
  - Make carbon source solutions as 10% (w/v) solutions and filter sterilize through 0.22 μm filters.
  - Use solutions at 10 mL per liter of medium.
- NH_4Cl is required at a final concentration of 0.2% (w/v) if nitrogen is not available from the carbon source. Of the carbon sources listed above, sodium benzoate, p-hydroxybenzoate, sodium succinate, and trehalose require NH_4Cl.
  - NH_4Cl: Add 10 mL of 20% (w/v) solution per liter of medium
- Anaerobic enrichment (optional) uses glycerol and NH_4Cl as carbon and nitrogen sources and requires KNO_3 as an electron acceptor.
  - KNO_3: Add 25 mL of a sterile 40% (w/v) solution per liter of medium.
  - Glycerol: Add 10 mL of a sterile 50% (v/v) solution per liter of medium.
  - NH_4Cl: Add 10 mL of 20% (w/v) solution per liter of medium**.**
- For solid media, add 15 g of agar per liter of medium and autoclave. Allow media to cool to ~50°C and add appropriate filter sterilized carbon (and nitrogen) source and dispense to Petri dishes.

TSA + Starch

- This medium is used to test for the ability to produce the enzyme α-amylase.
- Add the following ingredients to 1 liter of distilled water:
  - TSA (Sigma-Aldrich Cat. No. 22091): 40 g
  - Starch (Sigma-Aldrich Cat. No. 33615): 2.0 g
- Autoclave to sterilize.

The following media are prepared as described by the manufacturer.

Medium	Purpose	Sigma-Aldrich catalog number
Pseudomonas Isolation Agar (PIA)	isolation of *Pseudomonas* on solid media following enrichment	17208
Nitrate broth	test for the ability to reduce nitrate	72548
Pseudomonas F agar	test for fluorescein production	P1852
LB agar	routine culturing	L3147
LB broth	routine culturing	L3522

Molecular Biology Reagents

Genomic DNA Extraction

- Genomic DNA is isolated and purified using the DNeasy Blood and Tissue kit from Qiagen (Cat. No. 69506) according to the manufacturer's instructions for use with Gram negative bacteria.

PCR Reagents

- *Taq* DNA Polymerase (Invitrogen Cat. No. 10342-020)
  - The supplier includes the buffers for the PCR reaction**,** 10X PCR buffer and 50 mM $MgCl_2$.
- Custom Primers. Use at a final concentration of 1 µM. (**Dewhirst, F. E., T. Chen, J. Izard, B. J. Paster, A. C. R. Tanner, W.-H. Yu, A. Lakshmanan, and W. G. Wade.** 2010. The human oral microbiome. J Bact. **192**(19): 5002-5017.)

- Forward Primer: 5'-GAG TTT GAT YMT GGC TCA G-3'
- Reverse Primer: 5'-GAA GGA GGT GWT CCA DCC-3'

- dNTPs (100 mM, Invitrogen Cat. No. 10297-018). Use at a final concentration of 250 µM each dNTP.

Gel Electrophoresis

- 1.0% agarose (w/v) gel in 1X TAE
- Sample buffer is EZ-Vision One from Amresco (Cat. No. N472-Q-0.5ML). This product does not contain ethidium bromide; there are no special handling or disposal requirements.
- Molecular weight standard, e.g., 2-Log DNA Ladder from New England Biolabs (Cat. No. N3200S)

PCR Purification

- QIAquick PCR Purification Kit from Qiagen (Cat. No. 28106).

Cost estimate for molecular biology reagents.

	Units	Unit list Price ($)	Cost for 20 students ($)	Cost for 50 students ($)	Cost for 100 students ($)
Taq polymerase Invitrogen, 10342-020	500 U (100 µl)	206	206	206	412
dNTPs Invitrogen, 10297-018	4 X 250 µl	259	259	259	259
Custom primers (2) Life Technologies	pmol	10	10	10	10
QIAquick PCR purification kit Qiagen, 28104	50	108	108	216	324
DNeasy blood and tissue kit Qiagen, 69504	50	155	155	310	465
EZ-Vision One Amresco, N472-Q-0.5ML	0.5 ml	25	25	25	25
2-Log DNA Ladder	0.1 ml	54	54	54	54

New England Biolabs, N3200S					
Total			817	1080	1574

Materials Required, by Laboratory Period

Lab 1

- PMM mineral base, 25 ml per aerobic enrichment
- Carbon sources (10% w/v) and Nitrogen sources (10% w/v) as needed
- 250 ml flask for each aerobic enrichment
- 60 ml stoppered bottle for each anaerobic enrichment (optional)
- small graduated cylinders to measure 25, 60 ml
- Pasteur pipettes or droppers for adding carbon and nitrogen sources
- aluminum foil
- 0.5 g of soil per enrichment (supplied by students)

Lab 2

- PMM plate with the same carbon sources as in Lab 1, one plate of each carbon source per foursome
- Anaerobic jar with anaerobic gas generating sachet for anaerobic enrichments (optional)

Lab 3

- PIA plates, 10 per foursome

Lab 4

- Gram Stain reagents
- Filter Paper
- Oxidase reagent to test for the presence of cytochrome c oxidase
- LB plates, 7 per foursome

Lab 5

- Gram Stain reagents
- Filter Paper
- Oxidase reagent to test for the presence of cytochrome c oxidase
- LB plates, 5 per foursome
- LB broth, 5 ml per tube, 1 per student

Lab 6

- PMM plate with various carbon sources, 1 set per foursome
- *Pseudomonas* F agar plate, 1 per foursome
- TSA + Starch plate, 1 per foursome
- 3 LB plates per foursome
- Phosphate buffered saline, 5 ml per foursome
- 4 tubes TSB per foursome
- 4 tubes Nitrate broth per foursome
- Anaerobic Jar and gas generating sachet (optional)

Lab 7

- Genomic DNA Extraction kit with reagents aliquoted for each foursome
- 6X EZ-Vision loading dye
- 1.0% Agarose gel (one lane per student) and 1X TAE running buffer
- Gel box with power supply
- Molecular weight standard for agarose gel (1 lane per gel)
- Iodine for TSA + starch plate (same as for Gram stain)
- Ultraviolet light source and UV protective glasses
- tube of 1 ml sterile distilled water (one per student)
- gelatin strip (Fisher Scientific Cat. No. NC9560006) (one per student)

- α-naphthylamine solution (0.5% N, N-dimethyl naphthylamine (Sigma-Aldrich Cat. No. D4011) in 5N acetic acid)
- zinc dust (Sigma-Aldrich Cat. No. 209988)

Lab 8

- PCR Reagents as listed above (*Taq* polymerase, primers, dNTPs, buffer) aliquoted for each foursome
- Ice bucket with ice for assembling PCR reaction (one per pair of students)

Lab 9

- 1.0% agarose gel, running buffer, molecular weight standards, gel box and power supply
- 6X EZ-Vision loading dye
- QIAquick PCR Purification Kit with reagents aliquoted for each foursome
- Materials required for preparing samples for DNA sequencing (consult with sequencing service).

Lab 10

- Laptop computer with internet access for analyzing DNA sequence and accessing the Ribosomal Database Project website

Appendix 2: Student laboratory manual.

Enrichment, Isolation and Identification of *Pseudomonas*

Pseudomonads are **chemoheterotrophic** bacteria that generate energy by oxidative phosphorylation. They are obligate respirers (all respire aerobically; some respire anaerobically as well). Ubiquitous in soil, pseudomonads are Gram negative, oxidase-positive and motile by means of one or a few polar flagella.

Pseudomonads can have both positive and negative impacts on our lives and our environment. Some cause disease. For example, *Pseudomonas aeruginosa* can cause respiratory and urinary tract infections in humans and is particularly troublesome in burn patients and victims of cystic fibrosis. *Pseudomonas syringae* causes a yellowing/chlorotic condition in lilacs that is accompanied by lesions. On the other hand, a key characteristic of pseudomonads is the large number of diverse organic nutrients that they can metabolize. The ability to utilize many different (and often unusual) carbon sources makes many pseudomonad species important in nutrient cycling as well as in bioremediation. In fact, one species, the former *Pseudomonas multivorans* (now called *Burkholderia cepacia*), received its original name because of its extraordinary nutritional versatility. This species is capable of utilizing over 100 organic compounds as a sole source of carbon.

In this exercise, you will enrich for, isolate, and characterize bacteria of the genus *Pseudomonas* from soil. You will apply nutritionally selective conditions to the soil samples to isolate pseudomonads, however, you will not be able to isolate these organisms directly from the environmental samples because their numbers are generally quite low relative to those of other organisms present. Direct application of these environmental samples to an agar medium will likely yield few to no pseudomonad colonies. You will therefore need to **enrich** for pseudomonads in liquid media, increasing their relative numbers, before you can obtain isolates on solid media.

A good microbial enrichment strategy will exploit a characteristic of the target organism as a means of selectively promoting the growth of that organism and hindering the growth of the vast numbers of undesired microbes. As noted above, *Pseudomonas* species exhibit nutritional versatility and can catabolize carbon sources that most organisms cannot. A rich medium or a minimal glucose medium would be a poor choice for enrichment, as these media would facilitate the growth of a broad spectrum of organisms. A minimal medium with an exotic carbon source like sodium benzoate is a better choice; it would restrict growth to only those with the enzymes required for benzoate catabolism, including representatives of *Pseudomonas*.

The **basal medium** for the enrichment is:

33 mM Na-K-phosphate, pH 6.8

0.05% $MgSO_4$

0.005% $FeCl_3$

0.0005% $CaCl_2$

There is neither an organic carbon source nor a nitrogen source present in this medium and it can more appropriately be called a **mineral base** because all of the components are inorganic. The enrichment medium is generated by adding a carbon and nitrogen source to this mineral base; variations will be performed by different students. Within the class, there will be two types of enrichment cultures: some incubated aerobically that contain an organic carbon source and nitrogen source, and the other incubated anaerobically that contains glycerol, KNO_3, and NH_4Cl.

The first two incubations in this process are the inoculation of soil in an enrichment broth followed by streaking on the same medium in a solid (plate) format. The resulting colonies, exhibiting a variety of morphologies, represent a subpopulation of what exists in the original soil sample. While this represents a good enrichment for *Pseudomonas* species, the next task will be to sift through the colonies to find those that meet the criteria for *Pseudomonas*: Gram negative, oxidase positive, motile paired rods. In addition, many *Pseudomonas* species are not susceptible to the antibiotic triclosan, enabling the use of a selective medium, *Pseudomonas* Isolation Agar (PIA). (See the description of the media used.)

Practical considerations dictate the order of testing. The small colonies growing on minimal enrichment media can't provide sufficient material for all of these tests. Sacrifice half of a colony to prepare a wet mount and view it using phase contrast microscopy. Candidates that exhibit motile rod-shaped cells will be streaked on PIA, using the remaining half of the colony as the inoculum. Once a pure culture has been attained, colonies can be tested for Gram and oxidase reactions.

Taxonomy

Prokaryotic **taxonomy** has traditionally involved the classification of organisms grown in pure culture on the basis of their phenotypes (e.g., Gram reaction, ultrastructure, carbon source utilization, susceptibility to antibiotics). In these exercises, you will use differential media to test for some properties that are common among *Pseudomonas* species. Some strains of *Pseudomonas* produce pyoverdin or fluorescein, a fluorescent yellow

siderophore (pronounced SID-er-o-for; a protein that stores Fe^{+3} in a soluble format); *Pseudomonas* F medium is depleted in iron, inducing the bacteria to produce this compound to scavenge any available iron. Visual examination of *Pseudomonas* colonies on this medium is an easy way to identify the presence of pyoverdin by its yellow fluorescence. Other species of *Pseudomonas* synthesize and secrete **α-amylase**, an enzyme that hydrolyzes starch, (a glucose polymer) to glucose monomers. Starch is too large to be transported into the cell, so α-amylase is secreted to generate smaller glucose molecules that can be imported easily. Decades of microbiology research have culminated in the development of differential media that assess the properties of organisms by their reactions with the media.

In the 1970s, Carl Woese pioneered **molecular taxonomy**, focusing on **ribosomal RNA**, a molecule found in abundance in every organism. Remember that there are three types of RNA; 16S rRNA is a structural component of the ribosome, it is not translated into protein. The 16S rRNA molecule is ~1500 nucleotides long and both its **primary** (sequence) and **secondary structure** are considered heritable traits. Although RNA is single-stranded, it folds to form secondary structures called **stem-loops (hairpins)**, and other structures like **bulges** and **knots**. While 16S rRNA molecules have a generally similar shape, changes in nucleotide sequence can result in lengthening or shortening hairpin stems, forming unpaired bulges, or changing interactions between complementary single-stranded loops. Closely related organisms have more similar sequences and therefore comparisons of entire rRNA sequences allow for a general scheme of prokaryotic classification.

The most direct way to identify and classify an organism is to determine the **DNA sequence** of the gene that encodes its 16S rRNA and to compare it to the sequences in public databases. Those sequences with the greatest similarity to the test sequence are those that derive from its closest relatives. Keep in mind that you will isolate and amplify the **DNA** that encodes the **ribosomal RNA**. As these two molecules have complementary sequences, the RNA sequence can be inferred from the DNA.

The Polymerase Chain Reaction (PCR)

The **polymerase chain reaction (PCR)** technique selectively amplifies a DNA fragment from a complex mixture: one gene among the thousands in a bacterial chromosome or one gene among the millions in an environmental sample. In this application of PCR, you will amplify the *Pseudomonas* 16S rRNA gene from total genomic DNA extracted from a pure culture. As you would expect from a reaction catalyzed by **DNA polymerase**,

PCR copies DNA, specifically, a small piece of DNA whose ends are delineated by the location of two specific **oligonucleotide primers** designed by the experimenter. These short ~20-30 base single-stranded DNA molecules will hybridize to unique sequences within the *Pseudomonas* rRNA gene.

To identify the *Pseudomonas* rRNA gene in a mixture of DNA molecules derived from the entire chromosome, it's necessary to design **oligonucleotide primers**. Examine the alignment of five bacterial rRNA genes shown on the next page. These sequences represent a small portion of the 16S rRNA-encoding gene in each of five organisms. The numbers to the right of each sequence represents a position number, where position 1 is the first base transcribed from the gene. Note that the sequences shown here occur more than 1000 bases into the gene. In this alignment format, an asterisk beneath a position indicates the presence of the same nucleotide at that position in every sequence in the alignment. Long stretches of continuous asterisks identify **conserved regions**, regions that will have the same sequence. Look at the sequence in the first line of the five sequences. Most nucleotides at that place in the rRNA gene have identical nucleotides. Look at the sequence in the second line of the five sequences. Few asterisks are present here, and hyphens appear in the sequences where the lengths have had to be adjusted to improve the alignment. A sparse distribution of asterisks indicates a **variable region** of the rRNA molecule, where the sequence is specific to a particular genus or species. This short region of the alignment has genus-specific and species-specific information. To get the sequence of these taxonomically important variable regions, they are amplified from genomic DNA by PCR using primers derived from the conserved regions, and then subjected to DNA sequence determination.

The polymerase chain reaction is comprised of three separate reactions performed sequentially and then repeated 20-30 times (cycles). It is a chain reaction because each cycle doubles the number of copies of the target DNA. This results from the double-stranded nature of DNA; each strand and its complementary primer serve as a substrate for DNA polymerase. A piece of DNA present once in a sample will be replicated in the first cycle yielding two copies; after the second cycle, four copies will be present, and after 30 cycles, 2^{30} copies (10^9 copies) could theoretically result.

In the first step of each cycle (**denaturation**), double-stranded DNA to be copied (the **template**) is heat denatured to single-stranded DNA. This requires temperatures of ~95°C. In the second step (**annealing**), the temperature is lowered to 50-55°C to permit the binding of the oligonucleotide primers to their complementary

sequences on the denatured template DNA. In the third step (**extension** or **elongation**), DNA polymerase catalyzes the formation of phosphodiester bonds between the heteroduplex and the individual dNTPs to synthesize the

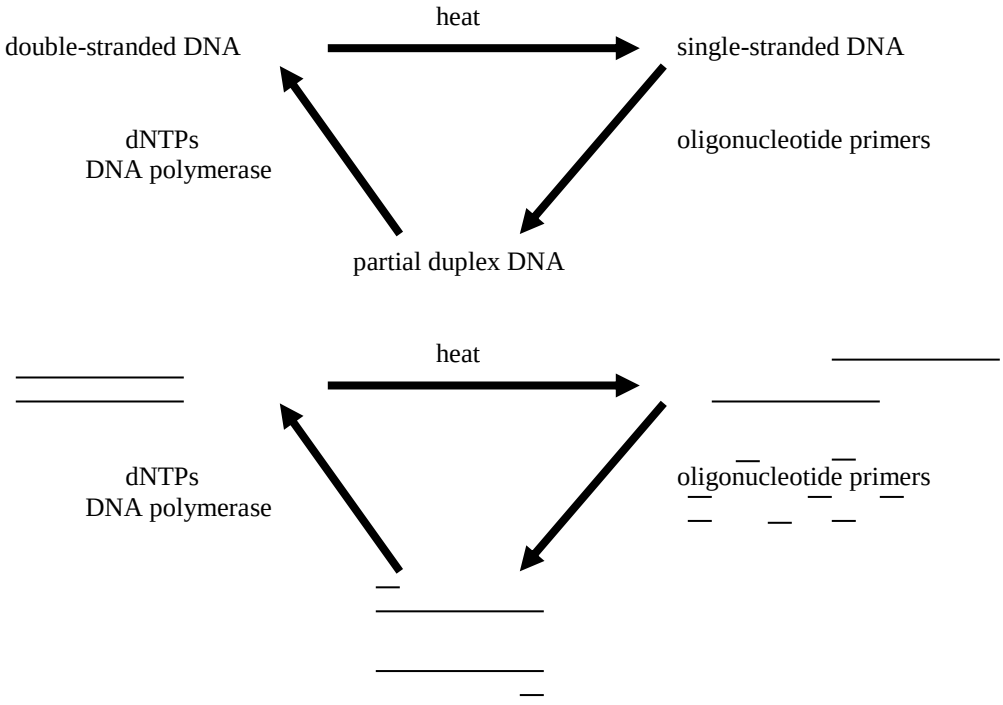
Multiple sequence alignment of five bacterial rRNA genes

```
Gallionella        CTGTCGTCAGCTCGTGTCGTGAGATGTTGGGTTAAGTCCCGCAACGAGCGCAACCCTTGT 1086
Pseudomonas        CTGTCGTCAGCTCGTGTCGTGAGATGTTGGGTTAAGTCCCGTAACGAGCGCAACCCTTGT 1085
Caulobacter        CTGTCGTCAGCTCGTGTCGTGAGATGTTGGGTTAAGTCCCGCAACGAGCGCAACCCTCGC 1033
Corallococcus      CTGTCGTCAGCTCGTGTCGTGAGATGTTGGGTTAAGTCCCGCAACGAGCGCAACCCTCGC 1117
Campylobacter      CTGTCGTCAGCTCGTGTCGTGAGATGTTGGGTTAAGTCCCGCAACGAGCGCAACCCACGT 1096
                   ************************************** **************  *


Gallionella        CATTAATTGCCATCATTCAGT--TGGGCACTTTAATGAGACTGCCGGTGATAAACCGGAG 1144
Pseudomonas        CCTTAGTTACCAGCACGTAATGGTGGGCACTCTAAGGAGACTGCCGGTGACAAACCGGAG 1145
Caulobacter        TGTTAGTTGCCACCATTTAGT--TGGGCACTCTAACAGGACTGCCGGAGTTAATCCGGAG 1091
Corallococcus      CTTTAGTTGCCA---CGCAAG--TGGAT-CTCTAGAGGGACTGCCGGTGTTAAACCGGAG 1171
Campylobacter      CATTAGTTGCTAACGGTTCGGC-CGAGCACTCTAATGAGACTGCCTTCG-TAAGGAGGAG 1154
                     *** ** * *           *    ** **    *******   *  **   ****

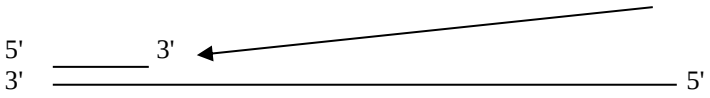

Gallionella        GAAGGTGGGGATGACGTCAAGTCCTCATGGCCCTTATGGGTAGGGCTTCACACGTAATAC 1204
Pseudomonas        GAAGGTGGGGATGACGTCAAGTCATCATGGCCCTTACGGCCAGGGCTACACACGTGCTAC 1205
Caulobacter        GAAGGCGGGGATGACGTCAAGTCCTCATGGCCCTTACAAGGTGGGCTACACACGTGCTAC 1151
Corallococcus      GAAGGTGGGGATGACGTCAAGTCCTCATGGCCTTTATGACCAGGGCTACACACGTGCTAC 1231
Campylobacter      GAAGGTGTGGACGACGTCAAGTCATCATGGCCCTTATGTCCGGGGCGACACACGTGCTAC 1214
                   ***** * *** *********** ******** ***      ****  *******  ***
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complementary DNA strand. The DNA polymerase used in PCR derives from a thermophile as such organisms have macromolecules that are stable at high temperatures. You will use ***Taq* DNA polymerase** (pronounced "tack") from *Thermus aquaticus* that works optimally at 72°C and can withstand temperatures of 95°C without loss of activity.

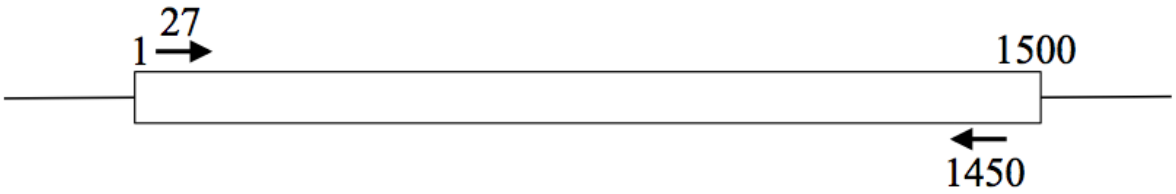


As with all *in vitro* reactions, a source of **energy** is needed for the reaction to proceed. Each of the three steps of PCR has specific energy sources: the denaturation step occurs by the energy from the **heat**; the annealing step is a **spontaneous** reaction (reversing the denaturation step); and the extension/elongation step is powered by the **dNTPs**, deoxynucleotide triphosphates, each of which contains a high energy triphosphate that provides the energy for its incorporation into a growing nucleotide chain.

Keep in mind the antiparallel nature of DNA – it is essential to know the 5' and 3' ends of each DNA strand and each oligonucleotide primer in order to understand PCR. Unless otherwise indicated, primer sequences are written in the 5' to 3' direction and the substrate for DNA polymerase is a 3' end.



The following diagram illustrates the 16S rRNA gene of *Pseudomonas* (~1500 bp) and the arrows indicate the approximate positions of the two primers that will be used for the PCR amplification. The primers are centered at ~27 bp and ~1450 bp in the gene.



The genomes of *Pseudomonas* species range from 5-7 **megabase pairs** (Mbp, 10^6 bp) and different species have 4-7 copies of the 1500 bp 16S rRNA. A quick calculation illustrates the power of PCR: If there are 4 copies of the rRNA gene (4 X 1500 bp = 6000 bp) in a 6 Mbp genome, then the rRNA genes constitute about 6,000 bp/6,000,000 bp or ~0.1% of the genomic DNA. Following PCR, the target (16S rRNA gene) constitutes >99% of the DNA present. It's important to remember that PCR is the **selective copying** of one region of the chromosomal DNA; there is no hydrolysis (cutting) of the DNA in this process.

Purification of Genomic DNA from *Pseudomonas*

The bacterial **genome** is defined as all of the hereditary material (DNA) contained within a given cell. This means that the bacterial genome is comprised of both the **chromosome** and any **plasmid** DNA that may be present in the cytoplasm. The size of a bacterial genome varies from species to species, but ranges from about 1.6 Mb (megabase pairs, or 1.6 X10^6 base pairs) to about 10 Mb. Free-living microbes with greater metabolic versatility (e.g., *Pseudomonas, Streptomyces*) tend to have larger genomes than intracellular parasites (*Mycoplasma*, *Rickettsia*). For many organisms, genomic DNA = chromosomal DNA, but for others genomic DNA = chromosomal DNA + plasmid DNA, and some organisms may have several different plasmids.

In this exercise, you will isolate and purify genomic DNA from a *Pseudomonas* isolate. Given the extreme **aspect ratio** (the ratio of length to width) of a bacterial chromosome – about 1 mm length vs 2 nm width – the molecule will break during isolation into pieces of about 20-30 kb (**kilobase pairs**, or 20,000-30,000 bp). You will therefore isolate and purify large fragments of the bacterial chromosome. As plasmids are generally much smaller, they may remain intact and circular through the purification. The procedure and materials that you will use are from a kit designed for exactly this purpose (genomic DNA purification from Gram negative bacteria), but it is imperative to review the chemical nature of the solutions provided in the kit and understand how they act to purify genomic DNA away from the other cellular components.

The bacterial culture is first centrifuged to separate the cells from the medium in which they were growing and the spent medium is discarded. The cells will be resuspended in lysis buffer, as this buffer is optimized for the initial reactions, and the growth medium is likely to have components that would interfere with these reactions. Lysis buffer contains **sodium dodecyl sulfate (SDS)**, a detergent that will solubilize membranes.

The second step in the purification of genomic DNA is the addition of **Proteinase K**, an enzyme that **hydrolyzes** most proteins (except, of course, itself – remember that enzymes are proteins). This enzymatic reaction is performed at 56°C, the optimal temperature for this enzyme. Recall the cell wall structure of a Gram negative bacterium – an outer membrane, a thin layer of peptidoglycan, and an inner membrane. This combination of SDS and Proteinase K will solubilize both membranes, and hydrolyze the peptide bonds in the peptidoglycan. Proteinase K is then still available to hydrolyze most of the cellular proteins once the cell has lysed. Remember that enzymes can perform many cycles of their catalytic activity – allowing the reaction to proceed for 30 minutes ensures that virtually all proteins are hydrolyzed.

The next steps in the purification of genomic DNA eliminate the remaining macromolecules until only the DNA remains. **RNase A** is an enzyme that hydrolyzes all classes of RNA – ribosomal RNA, transfer RNA, and messenger RNA. Note that RNase A is also a protein and is itself susceptible to hydrolysis by Proteinase K. However, large amounts of RNase A are added to the reaction and it proceeds at room temperature, far below the optimal temperature of Proteinase K (56°C). In addition, RNase A is very fast acting and will hydrolyze cellular RNA to short oligoribonucleotides almost instantly.

Guanidine hydrochloride, a chemical that **denatures** proteins, including Proteinase K and RNase A, is added next. The addition of **ethanol** to the mixture causes the DNA to precipitate from solution. This is a physical

change, in which the ethanol **dehydrates** the DNA and the individual DNA molecules have a greater affinity for one another than they do for the solvent, and the result is precipitation from solution. The entire mixture of precipitated DNA, hydrolyzed cellular components, denatured proteins, and soluble metabolite pools is applied to the **DNeasy mini column**. Only DNA binds to the column; all other components pass through, and the DNA has been isolated.

The remaining steps serve to purify the isolated DNA, by washing with ethanol-based buffers (AW1 and AW2). In each case, the DNA remains bound to the column matrix and contaminants are washed through into the collection tube. Centrifugation provides the force to wash the solutions over the bound DNA. The final step is **elution** of the DNA from the column matrix with an aqueous elution buffer (AE). For this final centrifugation, the column should be placed into a clean tube, as the wash-through will be the final pure DNA preparation.

Agarose Gel Electrophoresis

Agarose gel electrophoresis is used to separate DNA fragments so that they may be visualized. Agarose is a highly purified polysaccharide that is the primary component of the agar used in solid bacterial media. The migration of molecules through an electric field is dependent on a combination of size, overall charge, and shape. Because all molecules of DNA have an essentially identical charge to mass ratio (i.e., identical charge densities, or one phosphate per nucleotide), the rate of migration of linear molecules of DNA through an electric field is dependent almost entirely on the size of the molecule. As the phosphate groups in DNA carry a negative charge, DNA is a **polyanion** and migrates toward the **anode** (positive pole). For linear DNA, the mobility of a fragment is inversely proportional to the size of the molecule; that is, smaller fragments move faster. So if DNA samples are loaded into an agarose gel and allowed to migrate through an electric field, we can separate the molecules based on their size.

How can the fragments be visualized in the gel? DNA alone is invisible to the eye, so fluorescent dyes with an affinity for DNA are used. Upon excitation, fluorescent dyes will absorb the energy of UV light and emit lower energy visible light. This means that when you illuminate the gel with UV light after electrophoresis, the DNA fragments will fluoresce and will appear as bright bands on a dark background.

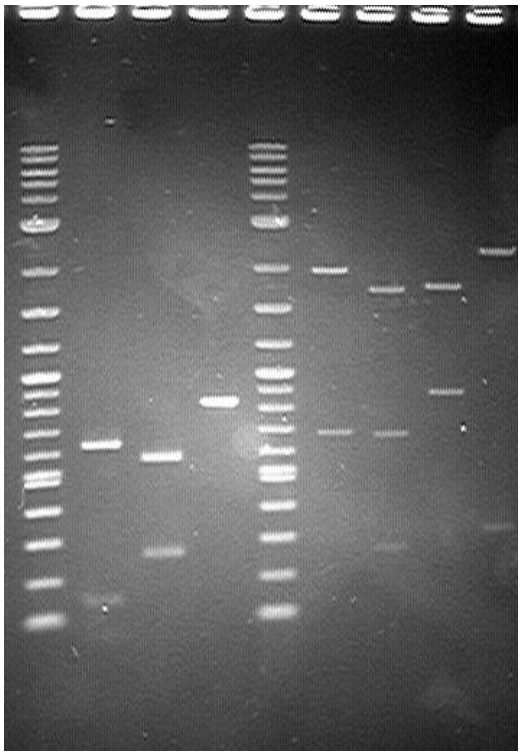
The agarose gel is a semisolid with the consistency of Jello, and a method of preparation that is similar to Jello. Agarose and gel buffer are mixed and heated until boiling to dissolve the agarose, cooled somewhat, and finally poured into a mold to provide the desired shape to the final product. Agarose gels are prepared in rectangular

trays, and while still molten, a “comb” is set into the gel to provide spaces for applying the nucleic acid samples. After the gel has solidified, the comb is removed and a row of wells (spaces in the gel) remains.

DNA samples are mixed with sample buffer prior to loading on the agarose gel. Sample buffer is a mixture of several dyes, including a visible dye to track the progress of the electrophoresis, and a DNA-binding dye that will be detectable under ultraviolet illumination. After samples have been loaded into the wells, an electric field is applied, and the samples are separated in 30-60 minutes.

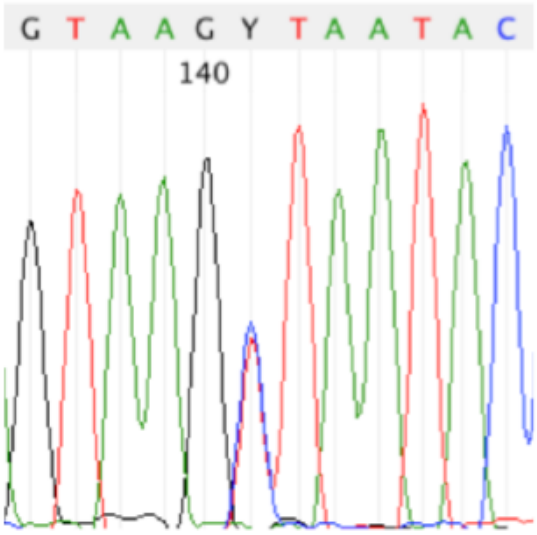
Images of agarose gels are taken under ultraviolet light, and the gels appear black with the bands of DNA glowing blue or orange (depending on which dyes are used). As most gel images are taken in black/white, the gel appears black and the bands white. A commercially prepared series of size standards should be run on every gel so that the sizes of the DNA fragments under observation can be estimated, and as a check on the electrophoresis and visualization processes.

In the image below, Lanes 1 and 5 (numbered from the left of the image) have the molecular weight standards, ranging from 10,000 bp (10 kilobasepairs or 10 kbp) at the top to 100 bp at the bottom. DNA samples in Lanes 2-4 and 6-9 may be sized by comparison to the standards. The wells into which the samples were loaded are visible at the extreme top of the image.



16S rRNA genes and the Ribosomal Database Project (RDP)

Because the 16S rRNA molecule is an essential structural component of the ribosome, and bacterial cells contain more than 10,000 ribosomes, most bacteria have multiple copies of the 16S rRNA gene in their genome. *Pseudomonas* genome sequences show that a single species has 4-6 genes that encode 16S rRNA, and these genes are not identical in sequence. They vary only by a few nucleotides (<10) over the entire 1500 bp of the gene. When you amplified the 16S rRNA gene from your *Pseudomonas* genomic DNA prep, you amplified all of the different versions (**alleles**) of the gene. (Remember that the primers bind to identical sequences.) The PCR product that you obtain is a population that includes the different alleles of the 16S rRNA gene in the genome. When we determine the DNA sequence of that PCR product, we should observe all of the different sequences present. However, since there are only a few differences, most of the DNA sequences will be the same, but you may find some instances where there are **single nucleotide polymorphisms**, or **SNP**s. These represent position in the DNA sequence where there are differences between individual copies.



For example, the image to the left shows the DNA sequencer output from a PCR product. Each nucleotide is labeled with a different colored marker and the detector recognizes that color and "calls" the base at that position. The peak stands out significantly from the baseline at most positions, but when the DNA from the PCR contains a SNP, the sequencer detects both nucleotides at a position. In this example, it is clear that position #141 has an equal mixture of C and T: both are detected with equal frequency and that frequency is about half of the other peaks. The interpretation is that there are two populations of DNA fragments within the PCR product, one with the sequence GTAAGCTAATAC and the other with the sequence GTAAGTTAATAC. Note that the software inserts a Y at this position, indicating a pYrimidine (C or T).

The existence of SNPs in the DNA sequences derived from *Pseudomonas* 16S rRNA gene raises a question about how organisms can be classified when multiple (slightly different) copies of the 16S rRNA exist in an organism. The Ribosomal Database Project (RDP) has a single entry (sequence) for each organism, and is therefore incomplete. However, it is still a useful tool in that it can identify close relatives. Within the genus *Pseudomonas*, there are more than 100 type strains that have been described and validated in the literature, and these have been divided into "subgroups" as documented by Mulet et al. 2010. (DNA sequence based analysis of the genus *Pseudomonas*. Environmental Microbiology 12: 1513-1530.)

Bacterial Growth Media

***Pseudomonas* Minimal (Broth or Agar) Media**

Designed for either the enrichment or isolation of pseudomonads, these media are minimal, defined, and selective. Their selectivity is determined primarily by the nature of the carbon and nitrogen source supplied, as the metabolic versatility of pseudomonads allows for the use of a variety of sources that are less commonly utilized by other microbes. Growth is limited to organisms that can utilize the C and N sources supplied. When these sources are combined with basic nutrients that are limited to the minimum needed to support the growth of pseudomonads,

the growth of many other organisms will be inhibited. In the broth that is used for enrichment cultures, this should reduce competition for nutrients and allow the number of pseudomonads to increase enough to allow for their ready isolation on agar. In addition, when special nutrient and anaerobic conditions are employed, the enrichment broth will allow for the growth of denitrifying bacteria. These bacteria can use nitrate (NO_3^-) rather than oxygen (O_2) as the terminal electron acceptor for the electron transport chain employed in anaerobic respiration. Note that in this particular medium, the nitrate does NOT serve as the source of nitrogen for growth.

Cultures grown in enrichment broth must be transferred to *Pseudomonas* minimal agar to obtain isolated colonies that can then be used to produce pure cultures. Once again, the nutritionally selective conditions present in these agar media should continue to favor the growth of pseudomonads over other organisms, making the isolation of pseudomonads more likely.

***Pseudomonas* Isolation Agar (PIA)**

Pseudomonas Isolation Agar (PIA) is a fairly rich, undefined, selective medium used for the isolation of members of the genus *Pseudomonas*. PIA provides a proteinaceous source of carbon and energy, peptone. It also contains Irgasan® (triclosan), a broad-spectrum antimicrobial agent that inhibits the growth a wide variety of organisms but does not select against *Pseudomonas*. This compound does not inhibit all other microbes, e.g., *Serratia*.

***Pseudomonas* F agar**

Designed for use with pseudomonads, *Pseudomonas* F agar is a rich, undefined, differential medium that contains tryptone and proteose peptone as proteinaceous sources of carbon, nitrogen, and energy that aid in pigment production. Glycerol is also provided as an alternative carbon and energy source. This medium is used to differentiate between pseudomonads that produce the yellowish fluorescent pigment pyoverdin (fluorescein) and those that do not. Phosphate is included in the medium to stimulate pyoverdin production as well as theoretically inhibit the production of pyocyanin, a blue-green pigment produced by some *Pseudomonas* species. The magnesium sulfate that is included in *Pseudomonas* F agar provides the cations needed for the activation of pyoverdin production.

Nitrate Broth

Nitrate broth is a fairly rich, undefined, differential medium with protein-based (beef extract, peptone) carbon and energy sources, as well as extra potassium nitrate and a Durham tube (a small, inverted tube) immersed in the broth. This medium is used to determine the ability of an organism to reduce nitrate to nitrite, nitrogen gas, or ammonia in the process of anaerobic respiration. Microorganisms that possess an electron transport system capable of reducing nitrate will produce one of the following byproducts in this medium: nitrite (NO^{2-}), ammonia (NH_3), nitrogen gas ($N_2\uparrow$), or nitrous oxide ($N_2O\uparrow$). See the interpretation protocol at the end of the manual.

Starch Agar

This rich, undefined, differential medium is used to determine if an organism has the ability to utilize starch as a carbon and energy source. The composition of the medium is similar to TSA, with tryptone (a digest of casein, a milk protein) and soytone (a digest of soybean meal) as the proteinaceous sources of carbon and energy as well as nitrogen and vitamins. The main difference in starch agar is the addition of a potato extract. Largely composed of starch (amylose), this addition makes the medium differential in nature. Starch molecules are too large to be absorbed by the cell and utilization of this large, complex carbohydrate requires the presence of the exoenzyme α-amylase, which hydrolyzes the starch into its component glucose molecules, allowing for their uptake into the cell. Because other carbon and energy sources are also supplied in starch agar, growth on this medium alone does not determine an organism's ability to utilize starch. The agar must first be flooded with an iodine solution that can turn starch present in the medium deep blue to black in color. If an organism growing on this medium utilizes the starch (that is, secretes α-amylase), the addition of the iodine solution will reveal a clear area in the vicinity of the streak where the starch has been degraded. This type of organism would be considered starch positive (+), which means that it can utilize starch as a carbon and energy source and produces α-amylase. However, if the organism does not produce α-amylase, the starch will remain in the medium, and the area immediately surrounding the organism will appear the same blue-black color as the rest of the medium after the iodine has been added. That organism would be considered starch negative (–).

Lab 1. Preparation of enrichment cultures for pseudomonads.

In the space provided below, record the assigned enrichment conditions for your pair.

_____

If you have been assigned an aerobic enrichment with a specific carbon/nitrogen source, follow the instructions below. If you have been assigned an anaerobic enrichment, see the instructions below and on the next page.

Preparation of aerobic enrichment cultures for pseudomonads.

• For your aerobic enrichment culture, measure 25 ml of enrichment medium mineral base and add it to a 250 ml flask. (It may have a slight precipitate – swirl to suspend.)

• The carbon and nitrogen sources that the class will use in these enrichments are listed in the table below.

Nutrient Sources Used for the Aerobic Pseudomonad Enrichment Cultures

Carbon Source (each at 10% w/v)	Volume of carbon source to add to 25 ml	**Nitrogen** Source (at 10% w/v)	Volume of nitrogen source to add to 25 ml
Na-benzoate	5 drops	NH_4Cl	5 drops
p-hydroxybenzoate	5 drops	NH_4Cl	5 drops
Na-succinate	5 drops	NH_4Cl	5 drops
Na-histidine	8 drops	(Na-histidine)	(no additional)
trehalose	5 drops	NH_4Cl	5 drops
arginine	8 drops	(arginine)	(no additional)
penicillin-G/ benzylpenicillin	8 drops	(penicillin-G/ benzylpenicillin	(no additional)

• Add the listed number of drops of your assigned carbon and nitrogen sources to the flask. Note that some compounds, specifically histidine, arginine, and penicillin-G, act as both carbon and nitrogen sources.

• Add 0.5 gram of soil to the flask. Record here the source of your inoculum: _____

• Cap the flask with two layers of aluminum foil and incubate at 30°C for a maximum of 2 days to minimize growth of other organisms.

Preparation of an anaerobic enrichment culture for denitrifying pseudomonads.

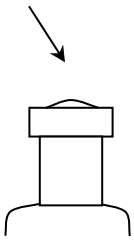
Nutrient Sources Used for the Anaerobic Pseudomonad Enrichment Cultures

Carbon Source	**Nitrogen** Source	Additional Ingredients
glycerol (50% v/v)	NH_4Cl (10% w/v)	KNO_3 (20% w/v)

• Label a reagent bottle and measure approximately 60 ml of enrichment medium mineral base. Add some to the reagent bottle, leaving about 2 cm of space from the base of the neck of the bottle.

• Set the bottle in half of a Petri plate to catch any medium that overflows. Carefully add 0.60 ml of 50% glycerol, 3.0 ml of 20% KNO_3 and 1 ml of 10% NH_4Cl to the to the solution in the bottle.

• Add 0.5 gram of dry soil. Record the source of your inoculum: _____



• Top off the bottle with enrichment mineral base so that a bead of liquid forms at the top, above the lip of the bottle (see illustration at left). Angle the glass stopper onto the bead of liquid and twist the stopper to close the bottle snugly (trying not to trap any air, and letting the excess liquid fall onto the Petri plate). Invert the bottle several times to mix the contents.

• Incubate the enrichment at 30°C for 2 days. Agitation is not necessary during this incubation.

Lab 2. Streaking pseudomonad enrichments for single colonies.

• Examine your enrichment culture. Does it appear different from when you began the incubation? How?

Beginning with Lab 2, students will work in groups of four to isolate *Pseudomonas* strains.

• Examine your two enrichment cultures (from both pairs in the group) under phase contrast microscopy for the presence of motile, paired rods.

• For each of the eight different carbon sources used in the enrichment cultures in the class, three-phase streak a loopful of the enrichment on a plate of minimal medium containing the identical carbon source. For example, streak a loopful of the sodium benzoate enrichment onto a plate of sodium benzoate minimal agar medium. Because your group of four performed only two enrichments, you will need to obtain samples of the other carbon source enrichments from other students in the class. Record the sources (i.e., "soil from OSU oval" not "Jane Smith") in the table on the next page.

• Label each plate completely, including both the carbon source included in the medium as well as the original source of inoculum (soil).

Carbon Source	Original Source of Inoculum
Na-benzoate	
p-hydroxybenzoate	
Na-succinate	
Na-histidine	
trehalose	
arginine	
penicillin-G (benzylpenicillin)	
glycerol (anaerobic)	

• Incubate all plates (except that derived from the anaerobic enrichment) at 30°C for 5 days. Store your broth enrichment cultures at 4°C.

• Incubate the glycerol-ammonium nitrate minimal medium plate in a GasPak® jar to generate an anaerobic environment. The GasPak jar will be incubated at 30°C for 5 days.

A "GasPak EZ Gas Generating Sachet," a gas-permeable pack that resembles a large tea bag, is used to establish the anaerobic conditions needed for the incubation of these cultures. Contained within the pack are inorganic carbonate, activated carbon, ascorbic acid, and water. The pack is removed from its airtight wrapper to activate its contents, and then placed directly in the incubation jar with the inoculated plates. The jar is sealed, and the activated carbon and ascorbic acid in the packet will reduce the oxygen content in the container by absorption and CO_2 will be generated from the carbonate. A strip that contains the oxygen indicator methylene blue is also added to the jar. If anaerobic conditions have actually been established after the jar has been sealed, the indictor strip should turn white, demonstrating that no gaseous oxygen is present. By varying the contents of the sachet used, this type of incubation jar can be used for the cultivation of anaerobic, microaerophilic or carbon-dioxide-requiring bacteria.

Lab 3. Presumptive identification of pseudomonads and streaking on a selective medium.

• Examine your minimal agar streak plates. Remember that these derived from liquid enrichments that were not pure cultures; you should expect to see a variety of colony morphologies on these plates. *Pseudomonas* colonies have entire, somewhat translucent edges. They are shiny and generally appear white to slightly off-white on these minimal media, although some may appear more colorful and opaque when grown on richer media. Avoid large, opaque colonies; these are likely *Bacillus*.

• Prepare wet mounts by setting up a "reservoir slide" with a few drops of water. For each wet mount, transfer a loopful of water from the reservoir to a portion of a microscope slide. Sample the colony with the sterile inoculating needle. Prepare wet mounts of small portions of your putative pseudomonad colonies for microscopic examination at 1000X, looking for paired rods. *Pseudomonas* strains are short rods and may be mistaken for cocci. (Motility may not always occur under these conditions.) If it appears to be a pseudomonad, you will need to use the remainder of the colony to three-phase streak onto a plate for isolation. Use the space below to make notes on your work.

• After confirmation by microscopic examination, streak putative pseudomonad colonies for isolation on *Pseudomonas* isolation agar (PIA). This medium is designed to select for the growth of *Pseudomonas* species by inhibiting others. It should therefore help narrow your search for putative pseudomonads.

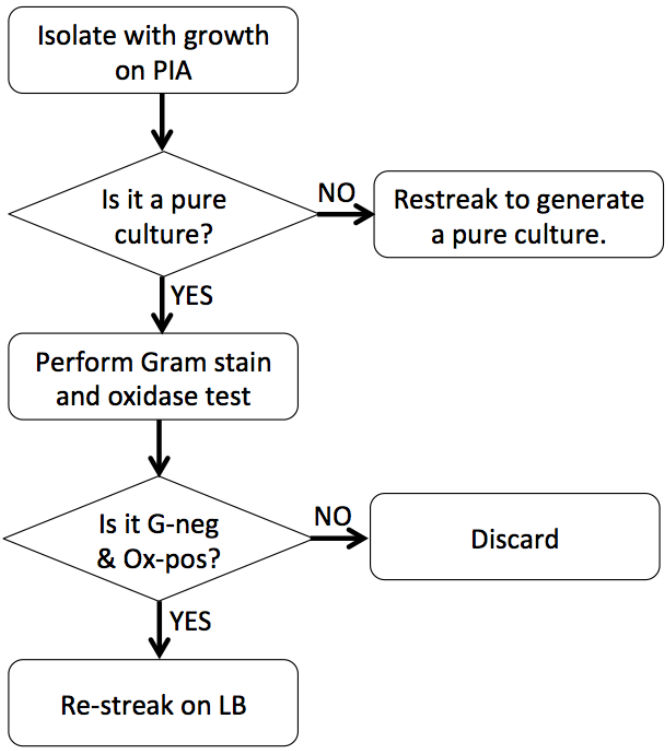
• Streak a total of ten PIA plates with different inocula. Diversity in this exercise comes from different soil samples, not from different carbon sources. Use the table provided below to help keep track of each isolate and assign each a number for future reference.

• Incubate the PIA plates at 30°C for 24 hr. Store minimal agar plates at 4°C.

Isolate #	Carbon Source	Original Source of Inoculum	Gram	Oxidase
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Labs 4 and 5. Confirmation of pure cultures.

Examine your streaks of presumptive *Pseudomonas* strains on PIA. Not all of the strains that you streaked are expected to grow, but your group should have at least one isolate per person. Of the isolates that did grow on PIA, examine the single colonies carefully to ascertain whether it is a pure culture. Subtle differences in colony morphology (opaqueness and pigmentation) may be apparent in the colonies, and it's quite possible at this stage in the enrichment to have a mixed culture. Follow the flowchart on the next page to guide each isolate's progress in the next two labs.



Do not proceed with the Gram and oxidase tests until you are certain that your culture is pure.

Each foursome must have a minimum of four pure cultures that meet all of the criteria for *Pseudomonas* species by the end of Lab 5. Restreak these putative *Pseudomonas* isolates onto separate LB agar plates. Transferring these cultures to an all-purpose medium like LB helps avoid any problems associated with carry-over from a selective medium like PIA that could affect growth and/or metabolism on the media that will be used to identify these microbes.

Lab 5. Start a broth culture of a *Pseudomonas* isolate.

Each student should inoculate a tube with a different isolate from your group's collection. All further experiments will be carried out with this broth culture. Write the number of your chosen isolate here:

_____

Inoculate with a single colony of *Pseudomonas*. Incubate at 30°C for 24 hours.

Lab 6. Prepare a glycerol stock of your isolate.

Bacterial strains are stored as glycerol stocks. Glycerol is a cryo-protectant and cultures mixed with glycerol can be frozen and stored indefinitely at -70°C. These can easily be prepared by mixing a liquid culture and glycerol and flash-freezing in liquid nitrogen. The strains are then revived as needed by inoculating fresh media with a small nugget of the frozen culture.

• Calculate how much 50% (v/v) glycerol should be mixed with your culture to get 1 ml of a suspension of cells in 15% (v/v) glycerol.

_____ µl of culture + _____ µl of 50% glycerol

• Label a screw cap tube, and in it mix sterile 50% glycerol and a sample of your overnight culture. Vortex to mix.

• Leave the mixtures in the designated rack for the TA to freeze.

• Save the remainder of your culture as it has two additional uses today.

Lab 6. Phenotypic Testing of *Pseudomonas*: Inoculating the test media.

• Completely label each of the following plates and divide each plate into four sections, one for each isolate in the group collection.

Pseudomonas minimal medium (PMM)

without a carbon source

PMM with glucose

PMM with histidine

PMM with trehalose

PMM with arginine

PMM with glycerol + nitrate

Pseudomonas F agar

TSA + starch

3 LB plates, labeled for incubation at 4°C, 30°C, and 42°C

• Prepare the cells from your broth culture for inoculation by "washing" them in saline. This eliminates any carry-over of the rich broth medium to the minimal media plates that might allow some growth. Transfer 1 ml of culture to a microcentrifuge tube and spin the tube in a microcentrifuge for 1 minute. Discard the supernatant in the beaker provided and then aseptically transfer 1 ml of saline to the tube. Vortex to resuspend the cells. Continue vortexing until there is no visible pellet and the suspension has turbidity similar to the starting culture.

• Inoculate each medium with your isolate in the appropriately labeled section on each plate.

• Label a tube of TSB for your isolate. This will be used to assay the presence of the enzyme gelatinase. Inoculate with a loopful of saline-washed cells. Vortex briefly to distribute the inoculum through the medium.

• Labe a tube of nitrate broth for your isolate. Submerged in each tube is a Durham tube. Inoculate with a loopful of saline-washed cells. Carefully roll each tube gently between your palms to distribute the inoculum through the medium and avoid introducing air into the Durham tube. Do not vortex these tubes.

• Place the LB plates labeled for incubation at 4°C and 42°C as indicated by your instructor.

• Incubate the glycerol + nitrate plate anaerobically at 30°C.

• Incubate the rest of the plates and tubes aerobically at 30°C.

Lab 6. Freeze cells for the purification of genomic DNA.

• Label two microcentrifuge tubes on the lid and transfer 1 ml of culture to each.

• Place the tubes (balanced!) in the microcentrifuge and spin at top speed for 1 minute. Discard the supernatants. Using a micropipettor, remove all traces of liquid.

• Place these pellets in the designated rack for freezing until the next class.

Lab 7. Purification of genomic DNA from a *Pseudomonas* isolate.

Each student will purify genomic DNA from his or her *Pseudomonas* isolate.

• Resuspend the pellet in 180 µl Lysis Buffer using a micropipettor. (Set the pipettor to 100 µl and use it as a "power washer" to resuspend the bacterial cells. As lysis buffer contains SDS, avoid introducing bubbles.)

• Add 20 µl of proteinase K. Use a micropipettor set to 100 µl to mix thoroughly by pipetting gently up and down a few times, and incubate at 56°C for 30 minutes. (During this incubation, continue with the other exercises.)

• Add 20 µl of RNase A (at 20 mg/ml), mix by vortexing, and incubate 2 minutes at room temperature.

• Vortex for 15 sec. Add 200 µl of guanidine hydrochloride to the sample and mix by vortexing.

• Add 200 µl ethanol and mix again by vortexing thoroughly. A white precipitate may form – this is the DNA.

• Pipet the mixture (including the white precipitate) into the DNeasy mini spin column placed in a 2 ml collection tube. Centrifuge at maximum speed for 1 min. Discard the flow through liquid in the beaker provided.

• Place the DNeasy mini spin column in a new 2 ml collection tube and add 500 µl Buffer AW1. Centrifuge at maximum speed for 1 min. Discard the flow through liquid.

• Place the DNeasy mini spin column in a new 2 ml collection tube and add 500 µl Buffer AW2. Centrifuge at maximum speed for 2 min. Discard the flow through liquid.

• Replace the DNeasy mini spin column in the same collection tube (no buffer is added this time). Centrifuge at maximum speed for 1 min.

• Place the DNeasy mini spin column in a clean microcentrifuge tube (NOT a collection tube) and add 100 µl Buffer AE. Incubate at room temperature for 1 min. Centrifuge at maximum speed for 1 min. The flow through liquid is your genomic DNA. Label the tube with the pertinent information.

Lab 7. Examine the genomic DNA prep and determine the presence of a plasmid.

One feature useful in classifying and identifying *Pseudomonas* strains is the presence of plasmid DNA in the genome. This can be assessed by visualizing the genomic DNA on an agarose gel. The chromosomal DNA has been fragmented to linear pieces of 20-30 kb during extraction and will appear as a single band larger than the largest band in the size standards. Plasmids will be smaller.

If you have never loaded samples on an agarose gel before, a gel will be available for practice.

• In a fresh microcentrifuge tube mix 10 µl of the genomic DNA and 2 µl of 6X EZ-Vision loading dye. Mix by setting your pipet to 6 µl and gently pipetting up and down.

• Load the entire 12 µl into a single lane of an agarose gel and record (both here and on the sheet accompanying the gel) in which lane you applied your sample. Gel # _____ Lane # _____

• The TAs will run the gels at 150V for about 1 hour and images will be posted online following class. See the diagram of the molecular size standards at the back of the manual.

• Record your results in the Summary Table at the back of the manual.

• Store your genomic DNA preparation at 4°C.

Lab 7. Phenotypic testing on *Pseudomonas* strains: observing and interpreting results.

• Observe and compare the relative growth for each isolate on the various minimal media and on LB agar incubated at 4°C, 30°C and 42°C. Record growth (+) or no growth (–) in the Summary Table on page 38. Use ± or multiple +, as necessary to indicate the relative amount of growth observed. (Some media require longer incubation and can be observed in the next class.)

• Test for α-amylase production. Flood the TSA + starch plate with iodine to test for the production of α-amylase. Observe the medium for the appearance of a clear zone around each streak (a positive reaction). You must observe the plate within 5-10 minutes of flooding, as the color will fade with time. Record a "+" for the utilization of starch/presence of α-amylase and a "–" for the lack of starch utilization/α-amylase production.

• View the isolates on *Pseudomonas*-F agar in the dark room under ultraviolet light to screen for the presence of pyoverdin (fluorescein), a fluorescent yellow pigment. Record a "+" for the presence of pyoverdin, and a "–" for its absence. Note that the fluorescent pigment is different from visible pigments.

• Test the isolates for gelatinase activity by vortexing the TSB culture and transferring three drops to a tube of sterile distilled water. Using flame-sterilized forceps, aseptically transfer a gelatin strip to each tube. Incubate at 30°C for 48 hours.

• Examine the growth of your isolates in nitrate broth. Consult the pages at the back of the manual for the details on interpretation of the results for this medium.

• Save the LB plate incubated at 30°C for use in a future lab (store it at 4°C).

Lab 8. Amplification of the 16S rRNA gene.

• Retrieve your genomic DNA preparation and label a small PCR tube.

• Assemble the following components in the PCR tube on ice:

primers mix		38 µl
buffer mix		8.5 µl
template DNA	(genomic DNA prep)	2 µl
Taq DNA polymerase		1 µl

• Mix the components by setting a micropipettor to 25 µl and gently pipetting up and down a few times.

• Place your tube in the thermal cycler. It has been set to a program called "ICE" that will maintain 0°C until all samples have been collected.

• The TAs will start the following program on the thermal cycler:

Step	Conditions	Number of cycles
1	92°C for 5 minutes	1
2	92°C for 30 seconds 55°C for 1 minute 72°C for 1 minute	30
3	72°C for 1 minute	1
4	4°C forever	1

• Store your genomic DNA preparation at 4°C.

Lab 8. Complete phenotypic testing.

• Retrieve your gelatinase test cultures and compare them to the control cultures. Clearing of gelatin from the plastic strip is an indication of gelatinase activity.

• Examine the pattern of growth on plates that were incubated for one week.

• Record (+) or (–) for each isolate in the Summary Table at the back of the manual.

Lab 9. Examination of *Pseudomonas* rRNA PCR products on an agarose gel.

Success of the PCR reaction is judged by running a small sample, typically 10% of the reaction, on an agarose gel and looking for the presence of a band of the expected size.

• Place your PCR tube in a de-capped microcentrifuge tube and spin briefly (a few seconds) to bring the liquid to the bottom of the PCR tube.

• Transfer 5 µl of your PCR to a fresh tube and add 5 µl of water and 2 µl of 6X EZ-Vision loading dye. Mix by setting your pipet to 6 µl and gently pipetting up and down.

• Load the entire 12 µl into a single lane of an agarose gel and record (both here and on the sheet accompanying the gel) in which lane you applied your sample. Gel # _____ Lane # _____

• The TAs will run the gels at 150V for about 1 hour and images will be posted online. See the image in the back of the manual for the molecular size standards used in this lab.

• If a band of the expected size has resulted from your PCR, purify the DNA product of the PCR.

Lab 9. PCR product purification.

Before its sequence can be determined, the PCR product (DNA) must be separated from the other components of the reaction. Materials for this exercise will be provided **after** the agarose gels are complete.

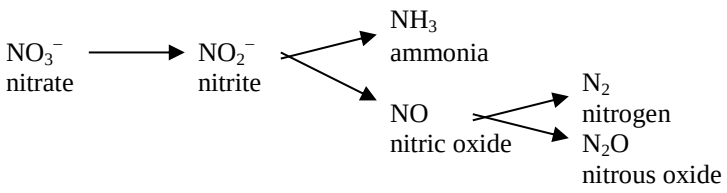
- Label two microcentrifuge tubes and a QIAquick (pink) spin column for your sample.
- Transfer your PCR (~45 µl remaining) to a 1.5 ml microcentrifuge tube and add 5 volumes (225 µl) of guanidine hydrochloride and mix well by vortexing.
- Apply this mixture to the QIAquick spin column that has been placed in a 2 ml collection tube. Centrifuge for 30-60 seconds.
- Discard the flow-through and replace the spin column in the collection tube.
- To wash, add 750 µl of ethanol wash buffer to the spin column and centrifuge for 30-60 sec.
- Discard the flow-through and replace the spin column in the collection tube.
- Centrifuge again (with no added buffer) for 2 minutes. (Residual wash buffer will inhibit the DNA sequencing reaction.)
- Discard the collection tube and place the pink spin column in a fresh (labeled) tube.
- To elute the DNA, add 30 µl of water to the center of the membrane, let the column stand undisturbed for 1 minute, then centrifuge for 1 minute.
- Discard the pink column and close the microcentrifuge tube. Check that this tube is properly labeled and place it in the designated rack to be submitted for DNA sequencing.

Lab 10. Download DNA sequence and compare it to the ribosomal database.

Classroom time will be devoted to ensuring that students get help with the computer programs needed to download DNA sequence, compare that sequence to the RDP, and to align DNA sequences.

Nitrate broth interpretation

The diagram below illustrates the pathways of nitrate metabolism.



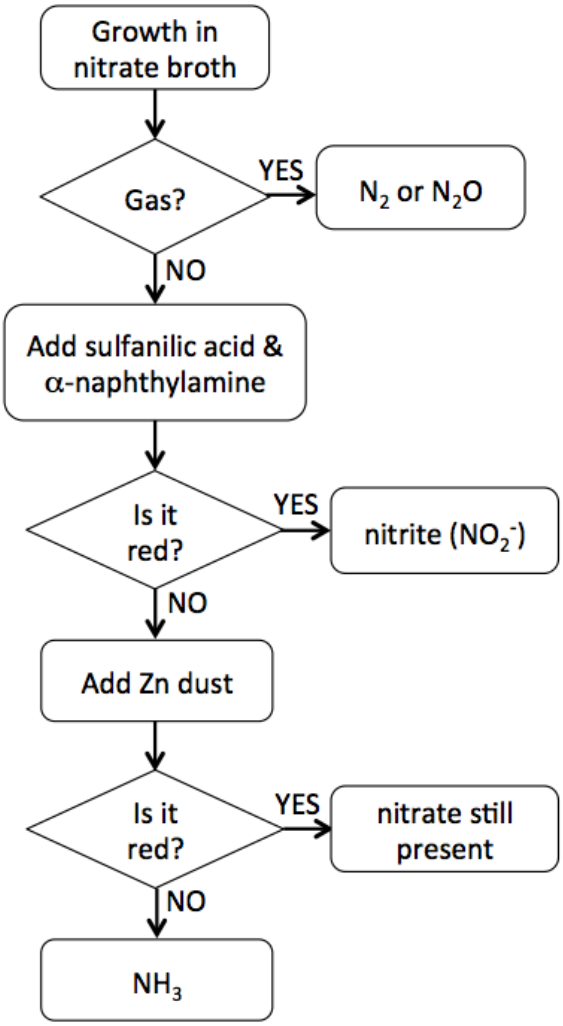
Growth in nitrate broth is used to determine the ability of an organism to reduce nitrate in the process of anaerobic respiration. Microorganisms that possess an electron transport system capable of reducing nitrate will produce one of the following byproducts in this medium: nitrite (NO_2^-), ammonia (NH_3), nitrogen gas ($N_2\uparrow$), or nitrous oxide ($N_2O\uparrow$).

The appearance of a bubble in the Durham tube is generally indicative of either $N_2\uparrow$ or $N_2O\uparrow$ production (a process known as **denitrification**, i.e., the removal of nitrogen from the medium). To either confirm that denitrification has occurred or, if no bubble is present, determine whether the nitrate in the broth has been reduced to nitrite, sulfanilic acid and α-naphthylamine solutions are added to the culture. In the acidic environment established by the sulfanilic acid, any nitrite that has been generated will react with the α-naphthylamine and produce a red color. The appearance of the red color is therefore a positive test for NO^{2-}, and no further testing is required because it has been established that nitrate reduction has occurred (i.e., nitrate has been reduced to nitrite). However, if this initial test is negative (no color change evident), it either means that the nitrate was not utilized at all or that it has been converted to something other than nitrite. To determine if the nitrate has not been utilized and therefore is still present in the medium, zinc is then added to the culture. Zinc will react with nitrate to produce a red color in the broth. The appearance of a red color at this point, then, indicates that the nitrate has not been utilized. The absence of a color change indicates that the nitrate is no longer present and must have been converted to $N_2\uparrow$, $N_2O\uparrow$ or NH_3. The lack of a gas bubble in the Durham tube indicates that the nitrate has been reduced to NH_3 and **ammonification** (a.k.a. nitrate mineralization) has occurred. This is summarized in the flow chart on the following page.

• First roll the tube between your palms to resuspend the nitrate broth culture, taking care to avoid introducing air bubbles into the Durham tube. Do NOT vortex. Inspect the Durham tube for the presence of gas bubbles indicating denitrification. Cultures exhibiting denitrification need not be tested further.

• To determine whether nitrate has been reduced to nitrite in the remaining culture, add 12 drops of sulfanilic acid solution and mix gently. Add α-naphthylamine solution dropwise to the tube and look for the formation of a red color in the medium. Red color within five minutes indicates the presence of nitrite and the culture need not be tested further if the reaction is positive.

• If no red color appears in the broth after the addition of the α-naphthylamine, use a spatula to add a very small amount of zinc dust to the tube. Zinc dust must be used in the hood. A pink color within five minutes indicates that nitrate is still present in the medium and there was no reduction of nitrate.



Summary Table

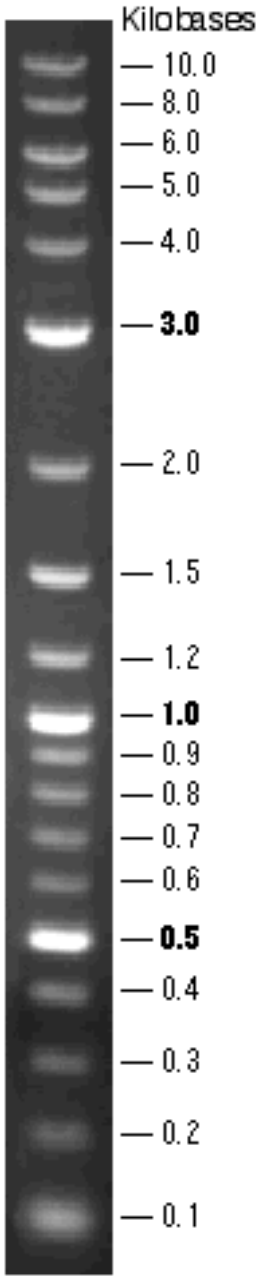
Plasmid in Genome								
Visible Pigmentation								
Fluorescent Pigmentation								
NO_3^- reduction (nitrate broth)								
NO_3^- electron acceptor (anaerobic plate)								
Gelatin Utilization								
Starch Utilization								
Trehalose Utilization								
Histidine Utilization								
Glucose Utilization								
Arginine Utilization								
Growth at 42°C								
Growth at 30°C								
Growth at 4°C								
Growth Without C								
Environmental Source								
Original C Source								
Isolate Number								

Molecular size standards for agarose gel electrophoresis

Molecular Weight Standard for agarose gels

2 log DNA ladder from New England Biolabs

Fragment	Size (bp)	Mass (ng)
1	10,002	40
2	8,001	40
3	6,001	48
4	5,001	40
5	4,001	32
6	**3,001**	**120**
7	2,017	40
8	1,517	57
9	1,200	45
10	**1,000**	**122**
11	900	34
12	800	31
13	700	27
14	600	23
15a	**517**	**124**
15b	**500**	**124**
16	400	49
17	300	37
18	200	32
19	100	61



DNA sequence analysis of the *Pseudomonas* rRNA gene

Formatting the DNA sequence for use in other programs

Most students should have several hundred bases of sequence. (If not, contact your TA.) Copy the text to a Word document. (Note on using Word for sequences – under the File menu, choose "Page Setup" and select the "landscape" format. Use the COURIER font and font size of 10-12.)

Before using your sequence, it must be edited, and this process requires judgment. Your sequence will have A, G, C, T, and N, and perhaps some others (Y, R). N represents "any" nucleotide and indicates that the software could not identify that residue unequivocally. The least reliable sequence is at the beginning and at the end (closest to the primer and near the resolving power of the detector). There will be a string of Ns at the beginning, usually 12-15. Delete those bases (Ns and other bases) from your sequence. For example, the following sequence

NNNNNNNNNNNNNNNNNCNCCTNCGGNAGGCAGCAGTGGGGAATATTGGACAATGGGCGAAAGCCTGATCCA

should be edited to

AGGCAGCAGTGGGGAATATTGGACAATGGGCGAAAGCCTGATCCA

to eliminate the region of sequence with Ns. It should NOT be edited as follows:

CCCTCGGAGGCAGCAGTGGGGAATATTGGACAATGGGCGAAAGCCTGATCCA

because these Ns are placeholders. Near the end of the sequence you will see an increasing frequency of Ns. This is a limitation of the technology, and you will probably need to eliminate dozens of bases. When the sequence is mostly Ns, delete; when it is about half Ns, save it.

Interpreting N in the middle of the sequence, as well as bases labeled R or Y (or other) requires examining the original data tracing from the sequencing reactions. Scan through the sequence, identify potential SNPs, and adjust the sequence as needed.

To prepare your data for a comparison, each sequence needs to be in an acceptable format, and a useful sequence format is called "**fasta**." Line 1 begins with the ">" symbol followed by a string of characters identifying the sequence (but no spaces) and a "return." The sequence then follows (upper or lower case) without gaps, as illustrated:

```
>myseq
ATGGCTAGCAAAGGAGAAGAACTTTTCACTGGAGTTGTCCCAATTCTTGTTGAATTAGATGGTGATGTTAATGGGCA
CAAATTTTCTGTCAGTGGAGAGGGTGAAGGTGATGCTACATACGGAAAGCTTACCCTTAAATTTATTTGCACTACTG
GAAAACTACCTGTTCCATGGCCAACACTTGTCACTACTTTCTCTTATGGTGTTCAATGCTTTTCCCGTTATCCGGAT
CATAT
```

Comparing your 16S rRNA sequence to the sequences in the Ribosomal Database (RDP)

The **Ribosomal Database** is exactly that – a compilation of 16S rRNA sequences that can be searched for matches or from which specific sequences can be downloaded. It is updated frequently to include sequences that have recently been deposited in public databases. In May 2014, there were 2,929,433 sequences in the database, and that number will certainly increase by the time you search for matches to your *Pseudomonas* rRNA sequence. The homepage for the Ribosomal Database is http://rdp.cme.msu.edu/.

From the RDP homepage you can search the database for those sequences most closely related to yours. Click on the Sequence Match (SEQMATCH) link and paste your DNA sequence into the box in fasta format. Below the box are some choices for which sequence you want to search within the database. Choose "Type" strains (well-characterized standard strains), Source "isolates," >1200 bp sequences (fairly complete sequences, not fragments), and "good" quality. Set KNN matches to "5." Click on "submit."

Under the "hierarchy view", click on "View selectable matches." You will be presented with a list of organisms clustered in taxonomic groups. As we're fairly certain that we are looking at *Pseudomonas* species, focus on those sequences (in the genus *Pseudomonas*). Each taxonomic group name is followed by three numbers in parentheses, for example, (0/20/116). The first number is the number of organisms that the program has selected as **good** matches for your query sequence; the second number is the number of **somewhat reasonable** matches; and the third number is the **total number** of sequences in that taxonomic category in the database.

Within each genus, there is a list of species; for each species, the following information is presented. The first number, in blue, is a clickable link to that organism's 16S rRNA gene sequence file in Genbank. The similarity score is highlighted in pink, although it is often not calculated and the words "not_calculated" will appear highlighted in pink. Highlighted in orange is the **S_ab score**, a measure of the degree of relatedness of your query sequence and the sequence listed. The higher the S_ab number, the more related the two sequences, and identical sequences have an S_ab score of 1.000. The final number shown is the number of unique common oligomers (not relevant for our purposes), followed by the genus and species name of the organism.

Downloading sequences from the RDP for use in other programs

Click on the blue number to open another window with the GenBank file for the organism's 16S rRNA gene. The fourth line lists the source organism, and published references describing the species of *Pseudomonas* are

listed next. Finally, the file contains the entire 16S rRNA gene sequence. Remember that this is the DNA sequence for a structural RNA – it will have T's not U's. The sequence is presented in groups of 10 nucleotides, 6 groups to a line, and each line is numbered. To convert this sequence to fasta format and to lose the gaps and numbers and lower case, go to a sequence conversion web site, e.g., http://www-bimas.cit.nih.gov/molbio/readseq/. Simply copy the DNA sequence (with gaps and numbers), paste it in the box, and click "submit." Copy the result to a Word file.

Aligning DNA sequences

The Clustal program accepts multiple sequences in fasta format and performs an alignment of those sequences. The program can be found at http://www.ebi.ac.uk/Tools/msa/clustalw2/. Copy the sequences from your Word document and paste them into the box. Accept all of the default settings and click on "run." Wait a few seconds for the program to run, and the output of the program (scroll down...) is an alignment. At each position where the sequences are identical, an asterisk appears below the alignment; at positions where the sequences differ, there is no symbol. This allows you to scan visually and locate differences between the sequences. The alignment can be copied and pasted into Word for presentation (use landscape orientation of the page). Note that if one of your sequences is upper case and the other is lower case, the program will not add asterisks.

Examining alignments and extracting meaning from them

The bacterial 16S rRNA gene is about 1500 bp long, and an alignment of several *Pseudomonas* genes will be extensive. Positions in the alignment that have asterisks are not overly informative, as those represent portions of the 16S rRNA that are identical in all *Pseudomonas* species. Look instead at the positions without an asterisk, where there are differences between/among the sequences.

Appendix 3: Instructor's guide.

Instructor's Guidelines and Formative Assessment Questions

These guidelines describe each day's activities and provide alerts to common pitfalls and incubation conditions, as well as formative assessment questions that may be used after the day's activity is complete.

Lab 1

Students begin this set of exercises working in pairs. Each pair is assigned a carbon/nitrogen source and students will bring their own soil sample (1/pair).

In a preliminary presentation, discuss aseptic technique, not used here. Aseptic technique is used with pure cultures and these are enrichments. Once a strain of *Pseudomonas* is isolated, aseptic technique is required to maintain a pure culture.

Aerobic *Pseudomonas* enrichments should be placed in a shaking incubator at 30°C for 48 hours. Anaerobic *Pseudomonas* enrichments should be placed in a 30°C incubator for 48 hours.

Lab 1 Formative Assessment Questions **(answers in bold**)

• Draw the chemical structures of the carbon and nitrogen sources used for enrichment. **(Structures may be found on the internet.)**

• Why is aseptic technique not used when handling the materials for the enrichment? **(See above explanation.)**

• Why is the anaerobic enrichment not placed in the shaking incubator**? (Broth cultures are incubated on a shaker to provide aeration, dissolving oxygen. The anaerobic enrichment uses nitrate instead of oxygen as the terminal electron acceptor.)**

Lab 2

In a preliminary discussion of the day's work, ask students to look at the enrichments and describe the differences from the previous class. Both enrichments are more turbid, and there will likely be gas bubbles present in the anaerobic enrichments. Denitrifiers generated N_2 gas; this is commonly observed.

Students will examine their enrichment under the microscope for paired rods. Motility is not always observed.

At this point, they will work in groups of four. Each foursome will get a set of Petri dishes, one plate with each of the carbon sources in use. The group of four will have two different enrichments, one from each pair, and will need to borrow from neighbors to get a sample of each carbon source used in the classroom. Students will streak a loopful from each enrichment culture onto the identical solid medium. Store the students' liquid enrichments at 4°C.

Aerobic enrichment plates should be sorted by carbon source and packed in plastic bags for incubation at 30°C. Most grow in 1-2 days but others (histidine and penicillin) need 5 days for colonies to appear. Anaerobic enrichments must be incubated in Gas Pak jars at 30°C for 5 days.

Lab 2 Formative Assessment Questions **(answers in bold**)

• Describe the cells observed in the enrichment broth. **(As primary enrichments, these are expected to have a variety of cell morphologies.)**

• What is the purpose of using the same enrichment medium in solid form? **(Colonies on plates derive from single cells, and generating a pure culture from a single cell is an essential part of isolating a strain.)**

Lab 3

The enrichment plates can be expected to show dozens of single colonies. As these are enrichments, not pure cultures, there should be a variety of colony morphologies. (See Figure 1A in the publication.) Students will attempt to identify colonies of *Pseudomonas* based on the colony description and cellular morphology, and then streak likely candidates on *Pseudomonas* isolation agar. Motility is often not observed on plates.

Record-keeping is essential here. Students will screen multiple colonies on the enrichment plates and need to correlate a wet mount with a colony. The entire colony can't be sampled to prepare the wet mount! For those that are good candidates to be *Pseudomonas* species, it should be given an identification number, the original source and enrichment carbon source should be noted, and the strain should be three-phase streaked on *Pseudomonas* Isolation Agar (PIA). Each group of four should aim to streak ten unique isolates, preferably from different soil sources, to ultimately end up with four isolates, one for each student.

All cultures will be maintained aerobically at this point even those originally from an anaerobic enrichment. Incubate PIA plates at 30°C for 1-2 days.

Lab 4

Expect that four to seven of the candidates streaked on PIA will grow. These are not necessarily pure cultures yet, and there may be more than one colony type present. (See Figure 1B and 1C in the original publication.) Before proceeding to the next steps, students must generate a pure culture, as judged by multiple identical colonies from a three-phase streak. Some cultures may appear pure on the PIA plate, and others may need to be re-streaked. Those who do not yet have pure cultures will finish early in Lab 4 and need the time in Lab 5 for the Gram and oxidase tests; those who have pure cultures will perform those tests in Lab 4 and will have extra time in Lab 5.

Some strains have visible pigmentation. We have observed colors from shades of brown/pink to pumpkin orange; these pigments are distinct from the fluorescent pigments.

We are defining *Pseudomonas* as Gram negative, oxidase positive, motile, paired rods. The initial screening should have identified motile, paired rods; as a first step, students will test Gram and oxidase reactions from those that are pure cultures. Strains that conform to the definition of *Pseudomonas* are streaked on LB.

As this incubation time corresponds with a weekend, and there are 5 days until the next meeting, plates are stored at 4°C for 3 days, then incubated at 30°C for 48 hours. Fresh cultures perform better in Gram and oxidase reactions.

Lab 4 Formative Assessment Questions **(answers in bold**)

- How do you judge if a strain is in pure culture? **(All single colonies should be identical in colony morphology.)**
- Why should the Gram and oxidase tests be performed on a pure culture? **(A mixed culture could give variable results.)**

Lab 5

This class is catch-up time for strains that were not pure after the first streak. Groups must have at least four different *Pseudomonas* isolates by the end of class; those with less than four can almost certainly find a group

with more than four who can share. This class marks the transition from group work to individual work; once each student chooses one of the isolates from the group collection, she or he is working as an individual.

Each student will inoculate a broth culture with a single colony of his/her isolate; all future work will be performed with this broth culture. It is often helpful to have a record of student names and the important information about their culture (soil source and enrichment; see Lab 10 instructions below). Students may be confused because the strain that they ultimately choose to characterize may not have derived from the soil sample and enrichment that they performed. The goal is to claim a unique isolate, not to follow through with their original enrichments.

Broth cultures are incubated on a shaker at 30°C overnight; some slow-growing strains require two nights.

Lab 6

The first use of this broth culture is to generate glycerol stocks for preservation, if desired.

The second use of the broth culture is to freeze two samples for later genomic DNA extraction: one to extract genomic DNA and a second as a spare if the first extraction fails. Supernatants should be removed from the pellets, and the pellets stored frozen. As labeling tubes legibly can be problematic, students are assigned unique seat numbers to avoid confusion over duplicate initials in the classroom.

The third use of the culture is to inoculate a variety of media for phenotypic testing. Most media can be incubated at 30°C for 5-7 days, to allow ample time for slow growth. *Pseudomonas* F agar, starch agar, TSB for gelatinase testing, and nitrate broth only need 2 days incubation. LB at 4°C and 42°C should have 7 days incubation.

Lab 6 Formative Assessment Questions **(answers in bold**)

• For what purpose is a glycerol stock used? **(Preservation and storage of the strain.)**

• Do you expect a strain of *Pseudomonas* to grow at all three temperatures, 4°C, 30°C, 42°C? **(Probably not, as that is a very broad range!)**

Lab 7

This class involves multitasking: extracting and purifying genomic DNA, while interpreting the results from the phenotypic testing and setting up additional tests.

The Qiagen DNeasy blood and tissue kit protocol for Gram negative bacteria has a 56°C incubation with Proteinase K that can be performed in any time frame from 20 minutes to several hours; this adds flexibility to class time management. Following the genomic DNA preparation, students load a sample on an agarose gel to confirm that their purification was successful and to examine the DNA for the presence of plasmids. If a sample "floats" out of the gel well, it has not been sufficiently separated from the ethanol wash and is therefore less dense; these preparations should be repeated using the second frozen cell pellet with more careful attention to the wash steps. Electrophoresis is performed after the class has ended and the image of the gel (annotated with lanes numbered) is posted online. Usually a few students have plasmid visible in the preparation. (See Figure 1D from the original publication.) A short required online quiz ensures that students review the image after class.

A common error is to load a sample of the genomic DNA on the gel, and then neglect to store the remaining DNA for future use; these tubes can often be recovered (by the student) from the biohazard trash bins where they were discarded. We require students to confirm that their sample has been saved before we sign them out after lab. Students whose preparation was unsuccessful must return to repeat it before the next class.

The following table indicates the expected reactions on the minimal and differential media used for phenotypic testing. Emphasize that we define *Pseudomonas* (genus) as Gram negative, oxidase positive, paired motile rods. The phenotypic data on the isolates will help to distinguish the species of *Pseudomonas*.

Medium	Purpose	What to look for
PMM without C source	negative control	no growth
PMM-glucose	positive control	growth
PMM-arginine	arginine as C/N source	growth/no growth
PMM-histidine	histidine as C/N source	growth/no growth
PMM-trehalose	trehalose as C source	growth/no growth
PMM-glycerol/nitrate	anaerobic growth	growth/no growth
TSA + starch	α-amylase production	starch iodine test
Pseudomonas F agar	fluorescent pigment prod.	view under UV for blue color
LB at 4°C	growth at 4°C	growth/no growth
LB at 30°C	positive control	growth
LB at 42°C	growth at 42°C	growth/no growth
gelatin strips	gelatinase production	strip visible/invisible
nitrate broth	denitrification	gas in Durham tube/reactions

Examination of the media for growth or no growth is easily interpreted. Positive control strains may be included, such as *Pseudomonas aeruginosa*, which should produce a positive reaction in the starch iodine and gelatinase tests and on *Pseudomonas* F agar. For interpretation of reactions in nitrate broth, control cultures of *P. aeruginosa* (reduces nitrate to ammonia), *E. coli* (reduces nitrate to nitrite), and *Alcaligenes faecalis* (does not reduce nitrate) are typically compared. (See for example, Leboffe, M.J. and B.E. Pierce. 2012. Microbiology Laboratory Theory and Application, 2nd ed. Morton Publishing, Englewood, Colorado).

As students work at different paces, some may need to store differential media for later evaluation. All media with cultures grown can be stored at 4°C, as these represent stable end-point reactions. The one exception is starch agar; it can be stored only if the iodine has not been added. Once the iodine has been added, the reaction must be evaluated within a few minutes.

Cultures with gelatin strips added should be incubated at 30°C for two days.

Lab 7 Formative Assessment Questions **(answers in bold**)

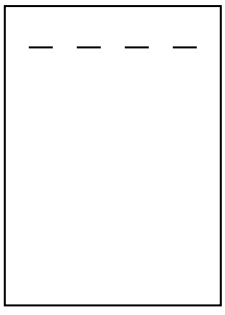
• Is aseptic technique necessary when extraction DNA from bacteria? Why or why not? **(It is not necessary. Although a few airborne contaminants may enter the culture during the purification, the presence of that DNA is insignificant.)**

• What's the difference between hydrolyzing a protein and denaturing a protein? **(Hydrolysis results in amino acids and short peptides; denaturation results in an intact polypeptide chain that is unfolded. Alternately phrased, hydrolysis destroys primary structure; denaturation destroys secondary and tertiary structure.)**

• What charge does a DNA molecule carry? What portion of the molecule is charged? **(The phosphate group in each nucleotide is negatively charged.)**

• On this diagram of an agarose gel, indicate where you would attach the positive and negative electrodes for DNA separation.

(negative)



(positive)

Lab 8

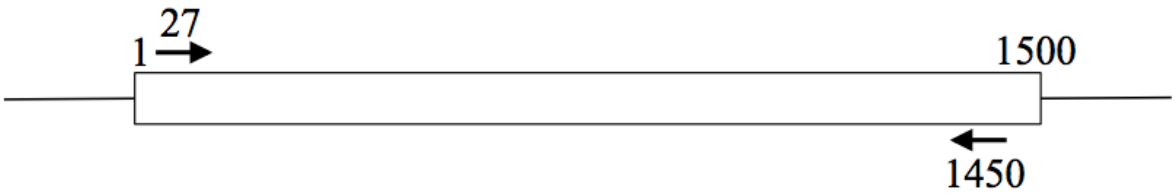
This lab also requires multitasking, to complete the evaluation of the phenotypic testing and to set up the PCR to amplify the 16S rRNA genes. In the pre-lab discussion, emphasize that the chromosomal DNA was not purified as one large circle but is in fragments of 20-30 kb; these are sufficiently long to act as the template for PCR.

For many students, the assembly of the PCR is their first molecular biology reaction, and students are not inspired by moving small volumes of clear liquids. Careful labeling of ingredients is critical. We pre-mix both primers and water as the "primer mix," and the buffer, Mg^{+2}, and dNTPs as the "buffer mix" so that students only have to measure four ingredients (these two mixes, their genomic DNA, and *Taq* polymerase). Set the thermal cycler to hold at 4°C as students place their reactions in it; once all tubes are prepared, the program can be initiated. Store PCRs and the remaining genomic DNA at 4°C or in the freezer until the next class.

Lab 8 Formative Assessment Questions **(answers in bold**)

• What ingredients must be in the buffer mix for PCR? **(buffer and dNTPs)**

• If the amplification is successful, what size dsDNA product would you expect from the PCR? **(1450 – 27 = 1423)**



• Would you expect the region around position 27 of the *Pseudomonas* 16S rRNA gene to be very similar, somewhat similar, or not at all similar between *Pseudomonas aeruginosa* and *Pseudomonas cepacia*? **(very similar)**

Lab 9

In this class, students prepare samples of their PCR for agarose gel electrophoresis. A common error is to add the sample dye to the entire reaction instead of removing a portion. This is not a fatal error, as the sample can be applied to the gel, and the remaining dye will be removed by the purification performed prior to DNA sequencing. Gels can be run quickly, as you are looking only for the presence of a product. Images are posted online following class, but students are notified immediately of the results. Those with successful reactions will

purify the product for DNA sequencing. Those with unsuccessful reactions need to repeat; it is virtually always successful the second time. The repeat reactions are assembled during this class. The genomic DNA preps are not needed unless a student has to repeat the PCR; these should be kept in storage until and if needed, as students can confuse them with the PCR products (both are small tubes of clear liquid).

Samples for DNA sequencing are submitted to a DNA sequencing facility immediately following class and the results are available in just over 24 hours. Either one or both primers may be used for DNA sequencing. Typically at least 800 bp of good sequence is obtained; this is sufficient for classification using the RDP. Using both primers will require finding the overlap in the sequence from the reverse complement. Although it is possible to perform, students may find this level of DNA sequence analysis confusing and distracting from the overall goal of classifying their *Pseudomonas* isolate.

Lab 10

Students download their *Pseudomonas* isolate's 16S rRNA gene sequence from the DNA sequencing facility website, edit the sequence, and use it to query the Ribosomal Database Project (RDP). They identify the closest relatives in the RDP, and align their isolate's sequence with those of its closest relatives using Clustal.

The challenge to editing the raw sequence is to remove the Ns that derive from sequence imprecision at the start and end of the sequence, but not those that indicate heterogeneity in the sequence. Certain single letters in the DNA code (Y, M, R, N) require that a student examine the original tracing and interpret the sequence in context.

The RDP results will vary with the length of the DNA sequence derived from a student's PCR product, and most of the sequence variation lies in the 5' portion of the 16S gene. It's useful to remind students that although the RDP contains a single DNA sequence for an organism, genome sequences illustrate that four or more copies with slightly difference sequences are present in most species of *Pseudomonas*.

Clustal is used to align the sequences from the RDP with the student sequence. All sequences entered must be given a unique name. Sequences downloaded from the GenBank files at the RDP must be converted to upper case by using a sequence conversion program before they can be used in the Clustal programs.

Students who needed to repeat the PCR can run the reaction on a gel today and prepare it for DNA sequencing. These students are asked to use another student's DNA sequence to learn the computer analysis. By

the end of this lab class, students should have a familiarity with the use and interpretation of DNA sequence alignments.

Students whose DNA sequencing reaction is unsuccessful may be provided with the sequence from another student. Assume that another isolate in the classroom from the same source is likely to be the same organism, and using the information collected in Lab 5 (see above), assign a sequence.

Identification of an isolate as either distinct from known species or as a member of known species is more of an art than a science, and students should be made aware of the relevant issues. As noted in the student lab manual, strains of *Pseudomonas* encode multiple different 16S rRNAs, and short of genome sequencing, only the sequence of a single 16S rRNA-encoding gene will be determined and compared. A recent publication (**Kim, M., H.-S. Oh, S.-C. Park, and J. Chun**. 2014. Towards a taxonomic coherence between average nucleotide identity and 16S rRNA gene sequence similarity for species demarcation of prokaryotes. IJSEM 64: 346-351) proposes a threshold of 98.65% identity (or 20 differences between two 16S rRNA genes) for new species. Students will not have sufficient DNA sequence data to measure, and therefore must employ comparisons of easier-to-obtain phenotypic data.

Appendix 4: Sample data.

The raw sequence shows many Ns that must be edited out before using the sequence to query the RDP.

```
>raw
NNNNNNNNGNNCNGNNNNNNNNNNNNNNNNNNGNNGNNNNGNNNNNNTNNNNNNNNNNNGNNNNNNNNNNAGCNNNNCCG
CGNNNGNGNNNANNNTCTTCNNNNNNNTAAAGCACTTTAAGTTGGGAGGAAGGGCATTAACCTAATACGTTAGTGTTTTG
ACGTTACCGACAGAATAAGCACCGGCTAACTCTGTGCCAGCAGCCGCGGTAATACAGAGGGTGCAAGCGTTAATCGGAAT
TACTGGGCGTAAAGCGCGCGTAGGTGGTTTGTTAAGTTGGATGTGAAAGCCCCGGGCTCAACCTGGGAACTGCATCCAAA
ACTGGCAAGCTAGAGTACGGTAGAGGGTGGTGGAATTTCCTGTGTAGCGGTGAAATGCGTAGATATAGGAAGGAACACCA
GTGGCGAAGGCGACCACCTGGACTGATACTGACACTGAGGTGCGAAAGCGTGGGGAGCAAACAGGATTAGATACCCTGGT
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CCTGGGGAGTACGGCCGCAAGGTTAAAACTCAAATGAATTGACGGGGGCCCGCACAAGCGGTGGAGCATGTGGTTTAATT
CGAAGCAACGCGAAGAACCTTACCAGGCCTTGACATGCAGAGAACTTTCCAGAGATGGATTGGTGCCTTCGGGAACTCTG
ACACAGGTGCTGCATGGCTGTCGTCAGCTCGTGTCGTGAGATGTTGGGTTAAGTCCCGTAACGAGCGCAACCCTTGTCCT
TAGTTACCAGCACGTTATGGTGGGCACTCTAANGAGACTGCCGGTGACAAACCGGANGAAGGTGGGGATGACGTCAAGTC
ATCATGGCCCTTACNGCCTGGCTACACANNNCNNCNAGGNNGNNNNNAANCCGCNAGNNNNGNNNNTAANCCGCGAGTNN
NNAACTAAAGCCGCGAGNNGNNAACNNAANCCGNNGTTGNNNACTAAANCCGCGAGNGGNNCTAANNCGCGAGGTGGAAN
NNNAACCNNAGNGGNAANNNAACCCGNNNGTGANNNAANNNGNNNCGNNNNNACANNNNNNNNGNNNNNATNNNNNNNNN
NN

>edited
TAAAGCACTTTAAGTTGGGAGGAAGGGCATTAACCTAATACGTTAGTGTTTTG
ACGTTACCGACAGAATAAGCACCGGCTAACTCTGTGCCAGCAGCCGCGGTAATACAGAGGGTGCAAGCGTTAATCGGAAT
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ACTGGCAAGCTAGAGTACGGTAGAGGGTGGTGGAATTTCCTGTGTAGCGGTGAAATGCGTAGATATAGGAAGGAACACCA
GTGGCGAAGGCGACCACCTGGACTGATACTGACACTGAGGTGCGAAAGCGTGGGGAGCAAACAGGATTAGATACCCTGGT
AGTCCACGCCGTAAACGATGTCAACTAGCCGTTGGAATCCTTGAGATTTTAGTGGCGCAGCTAACGCATTAAGTTGACCG
CCTGGGGAGTACGGCCGCAAGGTTAAAACTCAAATGAATTGACGGGGGCCCGCACAAGCGGTGGAGCATGTGGTTTAATT
CGAAGCAACGCGAAGAACCTTACCAGGCCTTGACATGCAGAGAACTTTCCAGAGATGGATTGGTGCCTTCGGGAACTCTG
ACACAGGTGCTGCATGGCTGTCGTCAGCTCGTGTCGTGAGATGTTGGGTTAAGTCCCGTAACGAGCGCAACCCTTGTCCT
TAGTTACCAGCACGTTATGGTGGGCACTCTAANGAGACTGCCGGTGACAAACCGGANGAAGGTGGGGATGACGTCAAGTC
ATCATGGCCCTTACNGCCTGGCTACACA
```

The ">edited" sequence was used to query the RDP as described in the student lab manual. The top five sequences were (with S_ab scores):

0.965 *Pseudomonas fulva*

0.963 *Pseudomonas putida*

0.954 *Pseudomonas parafulva*

0.954 *Pseudomonas cremoricolorata*

0.954 *Pseudomonas taiwanensis*

The genes encoding 16S rRNA from the two closest relatives, *P. fulva* and *P. putida*, were aligned with the sequence of this isolate for comparison. Shortly after the beginning of the sequence obtained from the isolate, a cluster of differences among the three strains is apparent by the lack of asterisks beneath the aligned sequences.

```
CLUSTAL O(1.1.0) multiple sequence alignment

fulva        ------ATCCTGGCTCAGATTGAACGCTGGCGGCAGGCCTAACACATGCAAGTCGAGCGG
putida       AGTTTGATCCTGGCTCAGATTGAACGCTGGCGGCAGGCCTAACACATGCAAGTCGAGCGG
isolate      ------------------------------------------------------------


fulva        ATGACGGGAGCTTGCTCCTTGATTCAGCGGCGGACGGGTGAGTAATGCCTAGGAATCTGC
putida       ATGAGAAGAGCTTGCTCTTCGATTCAGCGGCGGACGGGTGAGTAATGCCTAGGAATCTGC
isolate      ------------------------------------------------------------


fulva        CTGGTAGTGGGGGACAACGTTTCGAAAGGAACGCTAATACCGCATACGTCCTACGGGAGA
putida       CTGGTAGTGGGGGACAACGTTTCGAAAGGAACGCTAATACCGCATACGTCCTACGGGAGA
isolate      ------------------------------------------------------------

fulva        AAGCAGGGGACCTTCGGGCCTTGCGCTATCAGATGAGCCTAGGTCGGATTAGCTTGTTGG
putida       AAGCGGGGGACCTTCGGGCCTTGCGCTATCAGATGAGCCTAGGTCGGATTAGCTAGTTGG
isolate      ------------------------------------------------------------


fulva        TGAGGTAATGGCTCACCAAGGCGACGATCCGTAACTGGTCTGAGAGGATGATCAGTCACA
putida       TGAGGTAATGGCTCACCAAGGCGACGATCCGTAACTGGTCTGAGAGGATGATCAGTCACA
isolate      ------------------------------------------------------------


fulva        CTGGAACTGAGACACGGTCCAGACTCCTACGGGAGGCAGCAGTGGGGAATATTGGACAAT
putida       CTGGAACTGAGACACGGTCCAGACTCCTACGGGAGGCAGCAGTGGGGAATATTGGACAAT
isolate      ------------------------------------------------------------


fulva        GGGCGAAAGCCTGATCCAGCCATGCCGCGTGTGTGAAGAAGGTCTTCGGATTGTAAAGCA
putida       GGGCGAAAGCCTGATCCAGCCATGCCGCGTGTGTTAAGAAGGTCTTCGGATTGTAAAGCA
isolate      ----------------------------------------------------TAAAGCA
                                                                *******


fulva        CTTTAAGTTGGGAGGAAGGGTTGTAGATTAATACTCTGCAATTTTGACGTTACCGACAGA
putida       CTTTAAGTTGGGAGGAAGGGCATTAACCTAATACGTTAGTGTTTTCACGTTACCGACAGA
isolate      CTTTAAGTTGGGAGGAAGGGCATTAACCTAATACGTTAGTGTTTTGACGTTACCGACAGA
             ******************** : **.. ******  *. :.**** **************

fulva        ATAAGCACCGGCTAACTCTGTGCCAGCAGCCGCGGTAATACAGAGGGTGCAAGCGTTAAT
putida       ATAAGCACCGGCTAACTCTGTGCCAGCAGCCGCGGTAATACAGAGGGTGCAAGCGTTAAT
isolate      ATAAGCACCGGCTAACTCTGTGCCAGCAGCCGCGGTAATACAGAGGGTGCAAGCGTTAAT
             ************************************************************

fulva        CGGAATTACTGGGCGTAAAGCGCGCGTAGGTGGTTTGTTAAGTTGGATGTGAAAGCCCCG
putida       CGGAATTACTGGGCGTAAAGCGCGCGTGGGTGGTTTGTTAAGTTGGATGTGAAAGCCCCG
isolate      CGGAATTACTGGGCGTAAAGCGCGCGTAGGTGGTTTGTTAAGTTGGATGTGAAAGCCCCG
```

```
                ***************************.*********************************

fulva           GGCTCAACCTGGGAACTGCATCCAAAACTGGCAAGCTAGAGTACGGTAGAGGGTGGTGGA
putida          GGCTCAACCTGGGAACTGCATCCAAAACTGGCAAGCTAGAGTACGGTAGAGGGTGGTGGA
isolate         GGCTCAACCTGGGAACTGCATCCAAAACTGGCAAGCTAGAGTACGGTAGAGGGTGGTGGA
                ************************************************************

fulva           ATTTCCTGTGTAGCGGTGAAATGCGTAGATATAGGAAGGAACACCAGTGGCGAAGGCGAC
putida          ATTTCCTGTGTAGCGGTGAAATGCGTAGATATAGGAAGGAACACCAGTGGCGAAAGCGAC
isolate         ATTTCCTGTGTAGCGGTGAAATGCGTAGATATAGGAAGGAACACCAGTGGCGAAGGCGAC
                ************************************************.*****

fulva           CACCTGGACTGATACTGACACTGAGGTGCGAAAGCGTGGGGAGCAAACAGGATTAGATAC
putida          CACCTGGACTGATACTGACACTGAGGTGCGAAAGCGTGGGGAGCAAACAGGATTAGATAC
isolate         CACCTGGACTGATACTGACACTGAGGTGCGAAAGCGTGGGGAGCAAACAGGATTAGATAC
                ************************************************************

fulva           CCTGGTAGTCCACGCCGTAAACGATGTCAACTAGCCGTTGGAATCCTTGAGATTTTAGTG
putida          CCTGGTAGTCCACGCCGTAAACGATGTCAACTAGCCGTTGGAATCCTTGAGATTTTAGTG
isolate         CCTGGTAGTCCACGCCGTAAACGATGTCAACTAGCCGTTGGAATCCTTGAGATTTTAGTG
                ************************************************************

fulva           GCGCAGCTAACGCATTAAGTTGACCGCCTGGGGAGTACGGCCGCAAGGTTAAAACTCAAA
putida          GCGCAGCTAACGCATTAAGTTGACCGCCTGGGGAGTACGGCCGCAAGGTTAAAACTCAAA
isolate         GCGCAGCTAACGCATTAAGTTGACCGCCTGGGGAGTACGGCCGCAAGGTTAAAACTCAAA
                ************************************************************

fulva           TGAATTGACGGGGGCCCGCACAAGCGGTGGAGCATGTGGTTTAATTCGAAGCAACGCGAA
putida          TGAATTGACGGGGGCCCGCACAAGCGGTGGAGCATGTGGTTTAATTCGAAGCAACGCGAA
isolate         TGAATTGACGGGGGCCCGCACAAGCGGTGGAGCATGTGGTTTAATTCGAAGCAACGCGAA
                ************************************************************

fulva           GAACCTTACCAGGCCTTGACATGCAGAGAACTTTCCAGAGATGGATTGGTGCCTTCGGGA
putida          GAACCTTACCAGGCCTTGACATGCAGAGAACTTTCCAGAGATGGATTGGTGCCTTCGGGA
isolate         GAACCTTACCAGGCCTTGACATGCAGAGAACTTTCCAGAGATGGATTGGTGCCTTCGGGA
                ************************************************************

fulva           ACTCTGACACAGGTGCTGCATGGCTGTCGTCAGCTCGTGTCGTGAGATGTTGGGTTAAGT
putida          ACTCTGACACAGGTGCTGCATGGCTGTCGTCAGCTCGTGTCGTGAGATGTTGGGTTAAGT
isolate         ACTCTGACACAGGTGCTGCATGGCTGTCGTCAGCTCGTGTCGTGAGATGTTGGGTTAAGT
                ************************************************************

fulva           CCCGTAACGAGCGCAACCCTTGTCCTTAGTTACCAGCACGTTATGGTGGGCACTCTAAGG
putida          CCCGTAACGAGCGCAACCCTTGTCCTTAGTTACCAGCACGTAATGGTGGGCACTCTAAGG
isolate         CCCGTAACGAGCGCAACCCTTGTCCTTAGTTACCAGCACGTTATGGTGGGCACTCTAANG
                ****************************************:*********** *

fulva           AGACTGCCGGTGACAAACCGGAGGAAGGTGGGGATGACGTCAAGTCATCATGGCCCTTAC
putida          AGACTGCCGGTGACAAACCGGAGGAAGGTGGGGATGACGTCAAGTCATCATGGCCCTTAC
isolate         AGACTGCCGGTGACAAACCGGANGAAGGTGGGGATGACGTCAAGTCATCATGGCCCTTAC
                ********************* ************************************

fulva           GGCCTGGGCTACACACGTGCTACAATGGTCGGTACAGAGGGTTGCCAAGCCGCGAGGTGG
putida          GGCCTGGGCTACACACGTGCTACAATGGTCGGTACAGAGGGTTGCCAAGCCGCGAGGTGG
isolate         NGCCTGG-CTACACA---------------------------------------------
                 ****** *******
```

```
fulva       AGCTAATCTCACAAAACCGATCGTAGTCCGGATCGCAGTCTGCAACTCGACTGCGTGAAG
putida      AGCTAATCTCACAAAACCGAACGTAGTCCGGATCGCAGTCTGCAACTCGACTGCGTGAAG
isolate     ------------------------------------------------------------
fulva       TCGGAATCGCTAGTAATCGCGAATCAGAATGTCGCGGTGAATACGTTCCCGGGCCTTGTA
putida      TCGGAATCGCTAGTAATCGCGAATCAGAATGTCGCGGTGAATACGTTCCCGGGCCTTGTA
isolate     ------------------------------------------------------------

fulva       CACACCGCCCGTCACACCATGGGAGTGGGTTGCACCAGAAGTAGCTAGTCTAACCTTCGG
putida      CACACCGCCCGTCACACCATGGGAGTGGGTTGCACCAGAAGTAGCTAGTCTAACCTTCGG
isolate     ------------------------------------------------------------

fulva       GAGGACGGTTACCACGGTGTGATTCATGACTGGGGTGAAGTCGTAACAAG----------
putida      GAGGACGGTTACCACGGTGTGATTCATGACTGGGGTGAAGTCGTAACAAGGTAGCCGTAG
isolate     ------------------------------------------------------------

fulva       ---------------------------
putida      GGGAACCTGCGGCTGGATCACCTCCTT
isolate     ---------------------------
```

Representative student data from phenotypic tests

Plasmid in Genome	-	-	-	-
Visible Pigmentation	+	-	+	+
Fluorescent Pigmentation	+	-	+	+
NO_3^- reduction (nitrate broth)	-	-	-	+
NO_3^- electron acceptor (anaerobic plate)	+	+	-	-
Gelatin Utilization	-	-	-	-
Starch Utilization	+	-	+	+
Trehalose Utilization	-	-	+	+
Histidine Utilization	-	+	+	+
Glucose Utilization	+	+	+	+
Arginine Utilization	+	+	+	+
Growth at 42°C	-	-	+	+
Growth at 30°C	+	+	+	+
Growth at 4°C	+	+	-	-
Growth without carbon source	-	-	-	-
Environmental Source	University Village	Browning Amphitheater	11th Avenue	Norwich Avenue
Original C Source	p-hydroxybenzoate	succinate	succinate	histidine

Appendix 5: Sample assignments and exam questions with answers and grading keys.

Although the initial isolation of *Pseudomonas* strains is performed in groups, all assignments are individual.

Assignment 1, due sometime in the time frame of Labs 4-7.

This assignment is intended to introduce students to the format of how new species of bacteria are described and reported in the International Journal of Systematic and Evolutionary Microbiology. A recent paper is provided to the students, along with a reading guide and some questions. The questions and answer key for a typical assignment is below.

Assignment 2, due sometime in the time frame of Labs 6-9.

This assignment is intended to familiarize students with querying the RDP by providing some sample partial 16S rRNA sequences. A typical assignment, including sample sequences, is below.

Assignment 3, due one week after Lab 10.

This completed assignment should read like a manuscript for IJSEM. The assignment and grading rubric are below.

Assignment 1. Instructions.

This homework assignment requires that you read the following paper:

"*Pseudomonas chlororaphis* subsp. *piscium* subsp. nov., isolated from freshwater fish" by Sarah E. Burr, Stefanie Gobeli, Peter Kuhnert, Elinor Goldschmidt-Clermont, and Joachim Frey. International Journal of Systematic and Evolutionary Microbiology, 2010, 60:2753-2757.

Reading guide

This type of paper is not divided into the traditional Abstract-Methods-Results-Discussion sections, but is rather a running description of the isolation and characterization of the organism, concluding with the formal description of the new species.

Read the paper carefully, skimming the sections related to the procedures that we will NOT use in our characterization of *Pseudomonas*, specifically the two paragraphs on page 2754 relating to "GC of cellular fatty acid methyl esters" and "DNA-DNA hybridization."

Assignment 1.

Name _____ **KEY** _____ Section _____

1. The authors describe the isolation of strain JF3835^T, and note that it had an identical sequence of its 16S rRNA as strain JF4413, which they had previously isolated. Are these two strains identical? Why or why not?

   No – Figure 1 shows that they differ in their ability to metabolize itaconate.

2a. Which genes were used in the multilocus sequence analysis?

   atpD, carA, recA

2b. These genes are referred to as housekeeping genes. Define the term "housekeeping genes" and write why are such genes used for this type of analysis.

   Housekeeping genes encode the proteins that are essential for a cell to live. They are used for this type of analysis because they are found in every organism with only minor sequence variations.

3. How would you distinguish a diffusible pigment from a non-diffusible pigment?

   By whether it diffuses into the medium or stays associated with the colony.

4. The length of the 16S rRNA molecule in *P. chlororaphis* is ~1530 bases, for all subspecies. The authors state that the 16S rRNA encoding gene of *P. chlororaphis* subsp. *auriantiaca* is 99.75% similar to the gene from strain JF3835^T. Calculate the number of nucleotides different in the 16S rRNA genes of these two strains.

   (1530)(0.9975) = 1526. The remaining 4 bases are different.

5a. What criteria are we using to define an organism as *Pseudomonas*? Would these two strains fit this definition?

   Gram negative, oxidase positive, motile paired rods

   Yes!

5b. Would all of the strains listed in Table 1 fit this definition?

   No – strains 12 and 13 are oxidase negative.

Assignment 2.

Name _____ **KEY** _____ Section/Room _____

This assignment uses the Ribosomal Database Project (RDP) website at http://rdp.cme.msu.edu/ and you may find it helpful to review the relevant pages from your lab manual.

From the home page of the RDP, click on "Browsers." (If you don't immediately go to a page labeled "Hierarchy Browsers – Start" click on "Start over" to get you there.) In the box of options, choose "Type" strains and "Isolates" sources; accept the remaining defaults (≥1200 bp, good quality, nomenclatural taxonomy). Click on the box to "Browse."

Note that the name of each domain and phylum is followed by three numbers in parentheses, e.g., (0/n/0). The first and third numbers reflect matches to specific queries, and in this portion of the database are recorded as zeroes; the second number indicates the number of sequences in the database from the indicated taxonomic group.

1a. Within the type strain dataset of isolates with good sequence data of >1200 bp, what percentage of the sequences from the domain Bacteria derive from *Firmicutes* ?

Firmicutes: 1953/ 9196 = 21.2 %

1b. Click on the phylum "*Firmicutes*" to expand the group: how many classes does this phylum include? (Unclassified groups should be ignored.) Which class has the greatest number of sequences in the RDP?

5 classes, Bacilli

Within the class "*Bacilli*", click on the family "*Bacillaceae 1*", then the genus "*Bacillus"*. Mouse-over the identification code for the species *Bacillus salarius* (the letter "S" followed by a 9-digit number) and click on the "Genbank" link. A new window will open with background information about the species, including the sequence of its 16S rRNA gene.

2a. Write the first ten nucleotides of the sequence **tgacgatggt**

2b. Is the sequence DNA or RNA? **DNA**

2c. What is the size (in bases) of the 16S rRNA from *Bacillus salarius*? **1380 bases**

2d. From where was this organism isolated? **Saline lake**

3. Each student is assigned a sequence to classify using this website. The sequences are in a Word document online and each sequence represents a portion of a 16S rRNA gene. Copy your assigned sequence from the Word document and paste it into the Sequence Match (SeqMatch) feature of the RDP. (If you don't immediately go to a page labeled "Seqmatch – Start" click on "Start over" to get you there.) Follow the procedure in your lab manual to learn about how to interpret the output of this program.

Genus of best match _____

Top three species of best matches _____

>Bacteroides /seq1

>Corynebacterium /seq2

>seq1
CGAAAGAANGATTAATACCCGATGGCATAGTTCTTCCGNNTGGTAGAACTATTAAAGAATTTCGGTCATCNATGGGG
ATGCGTTCCATTAGNNNGTTGGCGGGGTAACGGCCCACCAAGCCTNCGATGGATNGGGGTTCTGAGAGGAAGGTCCC
CCACNTTGGAACTGAGACACGGTCCNAACTCCTNCGNGAGGCAGCAGTGAGGAATNTTGGTCNATGGACGAGAGTCT
GAACCAGCCAAGTNGCGTGAAGGATGNNTGCCCTNTGGGTTGTNAACTTCTTTTATACGGGANTNAAGTGAGGCANN
TGTGCCNNTTTGTATGTACAGTATGAATNAGGATCGGCTAACACCGTGCCNGCNGCCNCGGTNATACGGAGGATCCN
AGCGTTATCCGGNTTTNTTNGGTTTNAAGGGAGCGTAGGCGGACGCTTAAGTCAGTTGTGAAAGTTTGCGGCTCAAC
CGTANNATTGCAGTTGATACTGGGTGTCTTGAGTACAGTAGAGGCAGGCGGAATTCGTGGTGTAGCGGTGAAATGCT
TAGATNTCACGAAGAACTCCGATTGCGAAGGCAGCNNNCTGGACTGTAACTGACGCTGATGCTCGAAAGTGTGGATA
TCNAACAGGATTNGATACCCTGGTAGTCCACACAGTNAACGATGAATACTCGCTGTTTGCGATNTACAGTAAGCGGC
CAAGCGNNAGCGTTNAGTATTCNACCTGGGGAGTACGCCGGCAACGGTGAAACTCAAAGGAATTGACGGGGGCCCGC
ACAAGCGGAGGAACATGTGGTTTNATTCGATGATACGCGAGGAACCTTACCCGGGCTTNAATTGCAACTGAATGATG
TGGAGACATGTCAGCCGCAAGGAAGNTGTGAAGGTGCTGCNTNGTTGTNNTCAGCTCGTGCCGTGAGGTGTNGGCTN
AAGTNCCATAACGAGCGCNACCCTT

>seq2
CTTGGATGGNGTTGTGGAAAGCTTNNGCGGTATGGGATGAGCTCGCGGCCTATCAGCTTGTTGNNGGGGTAATGGCC
TACCAAGGCGTCGACGGGTAGCCGGCCNGAGAGGGTGTACGGCCACATTGGGACTGAGATACGGCCCAGACTCCTAC
GGGAGGCAGCAGTGGGGAATATTGCACAAAGGGCGCAAGCCTGATGCANCGACGCCGCGTGGGAGATGACGGCCTTC
GGGTTGTAAACNCCTTTCGCAACCGACGAAGCTTTTGTGACGGTAGGTTGAGAAGAAGCACCGGCTAACTACGTACC
NNCAGCCGCGGTAATACGTAGGNTGCGAGCGTTGTCCAGAATTACTGGGCGTAAAGAGCTCGTAGGTGGTTTGTNGC
GTCGTTTGTGGAATACCGCAGCTTAACTGCGGGGTTGCAGGCGATACGGGCATAACTTGAGTGCTGTAGGGGAGACT
GGAATTCCTGGTGTAGCGGTGGAATGNNCAGATATCAGGAGGAACACCGATGGCAAAGGCAGGTCTCTGGGCAGTAA
CTGACGCTGAGGAGCGAAAGCATGGGGAGCGAACAGGATTAGATACCCTGGTAGTCCATGCCGTAAACGGTGGGCGC
TAGGTGTGAGTCCCTACCACGGGGTTCGTGCCGTAGCTAACGCNTTNAGCGCCCCGCCTGGNNNGTACGGCCGCAAG
GCTAAAACTCAAAGGAATTGACGGGGGCCCGCACAAGCGGCGGAGCATGTGGATTAATTCGATGCAACGCGAAGAAC
CTTACCTGGGCTTGACATACACCGGATCGCTGCAGNNATGTCAGCTTCCCTTTGTGGCTGGTGTACAGGTGGTGCAT
GGTTGTCGTCANNTCGTGTCGTGAGATGTTGGGTTAAGTCCCGCAAAGAGCGCAACCCTTGTCTTATGTTGCCAGCA
TTTGGTTGGGGACTCATGAGAGACTGCCGGGGTTAACTCGGAGGAAGGTGGGGATGACGTCAAATCATCATGCCCCT
TATGTCCAGGGCTTCACACATGCTACAATAGTCGGTACAACGCGTTTGCGAGCTCGTGAGGGTGGGCTAATCGCTGA
NAGCCGGTCGTAGTTCGNNNTGGGGTCTGCAACTCGACCCCATGAAGTCGGAGTCGCTAGTAATCGCAGATCAGCAA
CGCTGCGGTGAATACGTTNNCGGGCCTTGTACACACCGCCCGTCA

Assignment 3

Assignment for the *Pseudomonas* exercise.

Your report will take either of two forms:

(a) Based on the DNA sequence of the 16S rRNA gene and on the results of phenotypic testing, your report should present an argument detailing why your isolate should be classified as a known species of *Pseudomonas*. Your supporting evidence should include a comparison of the phenotypic properties that you measured and those described in the literature for the organism(s) with the greatest similarity from the RDP. The GenBank file will have the citations for each strain.

(b) Your report may advocate the creation of a new species of *Pseudomonas*. Propose a species name and describe the organism in the format exemplified by the *Pseudomonas* identification paper that you have been assigned to read. Specifically, it should be "Description of *Pseudomonas xxx* sp. nov." and it should address the difference(s) between this isolate and its closest match in the database.

• Summarize the work in isolating and identifying *Pseudomonas* strains.

  • What was the source of <u>your</u> isolate?

  • What criteria did you use to assign it to the genus *Pseudomonas*?

  (Because generating your group's collection involved borrowing from other groups who used different enrichments, your isolate may not have originated from your own enrichment. Be sure to specify the correct source and enrichment for <u>your</u> isolate.)

• Include the summary table (or similar format). A blank is online.

  • Summary table should be complete and logical. You may use the multiple lines to summarize the data from the close relatives of your isolate.

• Compare the DNA sequence from your isolate to the RDP.

  • Present an alignment of the partial rRNA sequences, highlighting relevant differences.

  • Format carefully on the page so that the reader can interpret the alignment.

• Identify the type strain in the RDP that is most similar to your isolate.

- Argue for a or b as described above. Do not argue both ways. You may suggest additional testing to resolve discrepancies.
- Consult the Mulet et al. paper and from Figure 3 of that paper, note which "group" your isolate belongs in.
- Presentation includes grammar, spelling, organization. Remember the relevant points of technical writing.

The grading rubric for this assignment is attached.

Grading rubric for Assignment 3

	Criteria				**Points**
	10	**5**	**3**	**0**	
Introduction		The concept and objectives of the experiment are described. The source of the organism and enrichment conditions are precisely defined.	Work is missing some concepts and objectives. The source of the organism and enrichment conditions are poorly defined.	Missing several concepts and objectives. Objective of work is unclear. The source of the organism and enrichment conditions are missing.	
Results	Summary table is consistent and complete. Sequence alignment annotated and formatted properly.	Summary table is mostly complete. Sequence alignment present and correct, but improper annotations and formatting.	Summary table is inconsistent or incomplete. Sequence alignment present and correct, but improper annotations and formatting.	Summary table is incomplete. Sequence alignment missing or inaccurate. Improper annotations and formatting on alignment.	
Discussion	*Pseudomonas* species identified citing phenotypic and genotypic results. Discrepancies are addressed with several additional tests that could be performed.	*Pseudomonas* species identified citing phenotypic and genotypic results. Discrepancies are mentioned but additional tests are lacking.	*Pseudomonas* species identified citing phenotypic and genotypic results but results are poorly integrated. Discrepancies are not adequately addressed.	*Pseudomonas* not identified. Conclusions undocumented. Discrepancies not addressed.	
Presentation		Use IJSEM format, write in third person, organization and grammar accurate. Genus and species in italics, literature properly cited. Works cited included.	Sections mislabeled, mostly written in third person. Organization and grammar has several mistakes. Genus and species not in italics. Literature improperly cited. Works cited included.	Poor organization, writing in first person, multiple grammar mistakes. Literature improperly cited or missing. No works cited included.	
				Total:	30 = max

Appendix 6: Pre/posttest to assess student learning.

For each question, circle the letter that precedes the best answer and put an "X" to indicate your confidence level for each answer.

1. The goal of a microbiological enrichment is to
    a. increase the nutrient composition of a target environment.
    b. increase the nutrient composition of a bacteriological medium.
    c. isolate a desired target organism from an environment in pure culture.
    d. increase the relative abundance of a desired target organism.

I am confident of this answer _____
I am guessing _____

2. Enrichment based on nutrient conditions is best carried out using ____ medium because ____ .
    a. rich; a wide variety of organisms will grow on a rich medium.
    b. rich; organisms will grow more rapidly on a rich medium.
    c. minimal; you can limit the growth of undesired competitors.
    d. minimal; the medium can be prepared with highly purified chemicals.

I am confident of this answer _____
I am guessing _____

3. Enrichment for organisms that normally grow in soil should be performed at
    a. 4°C, as soil organisms from Ohio can withstand the cold of an Ohio winter.
    b. 30°C, as that is most comparable to the normal environment of soil organisms in Ohio.
    c. 37°C, as that is the standard incubation temperature in the microbiology lab.
    d. 70°C, as that is most comparable to the normal environment of soil organisms in Ohio.

I am confident of this answer _____
I am guessing _____

4. If a facultative anaerobe is isolated from
    a. an anaerobic enrichment, it must always be cultivated anaerobically.
    b. an anaerobic enrichment, it can be cultivated aerobically.
    c. an aerobic enrichment, it can never be cultivated anaerobically.

I am confident of this answer _____
I am guessing _____

5. An organism can be judged to be in pure culture on the basis of a three-phase streak when
    a. at least a dozen single colonies all have the same morphology.
    b. at least two single colonies are present.
    c. the primary streak is a uniform color.
    d. at least a dozen single colonies are present, regardless of morphology.

I am confident of this answer _____
I am guessing _____

6. A typical bacterial genome has
    a. a single copy of the gene that encodes the 16S rRNA.
    b. at least a dozen identical copies of the gene that encodes the 16S rRNA.
    c. several copies of very similar but not identical genes that encode the 16S rRNA.

I am confident of this answer _____
I am guessing _____

7. The sequence of the 16S rRNA can be used to identify organisms
    a. to the level of species.
    b. to the level of genus.
    c. to the level of class.
    d. to the level of domain.

I am confident of this answer _____
I am guessing _____

8. Two individual microbes of the same species may differ in
    a. the presence/absence of pigmentation.
    b. the presence/absence of a secreted enzyme.
    c. the presence/absence of a metabolic pathway.
    d. all of the above.
    e. none of the above.

I am confident of this answer _____
I am guessing _____

9. The 16S rRNA molecule is present in a bacterial cell at an abundance of
    a. one molecule per cell.
    b. 40-50 molecules per cell.
    c. 500-700 molecules per cell
    d. 10,000-15,000 molecules per cell.

I am confident of this answer _____
I am guessing _____

10. The Ribosomal Database Project contains a collection of 16S rRNA sequences
    a. from bacterial type strains.
    b. from environmental isolates.
    c. from validly published species.
    d. both a and b.
    e. both a and c.
    f. all of the above.

I am confident of this answer _____
I am guessing _____